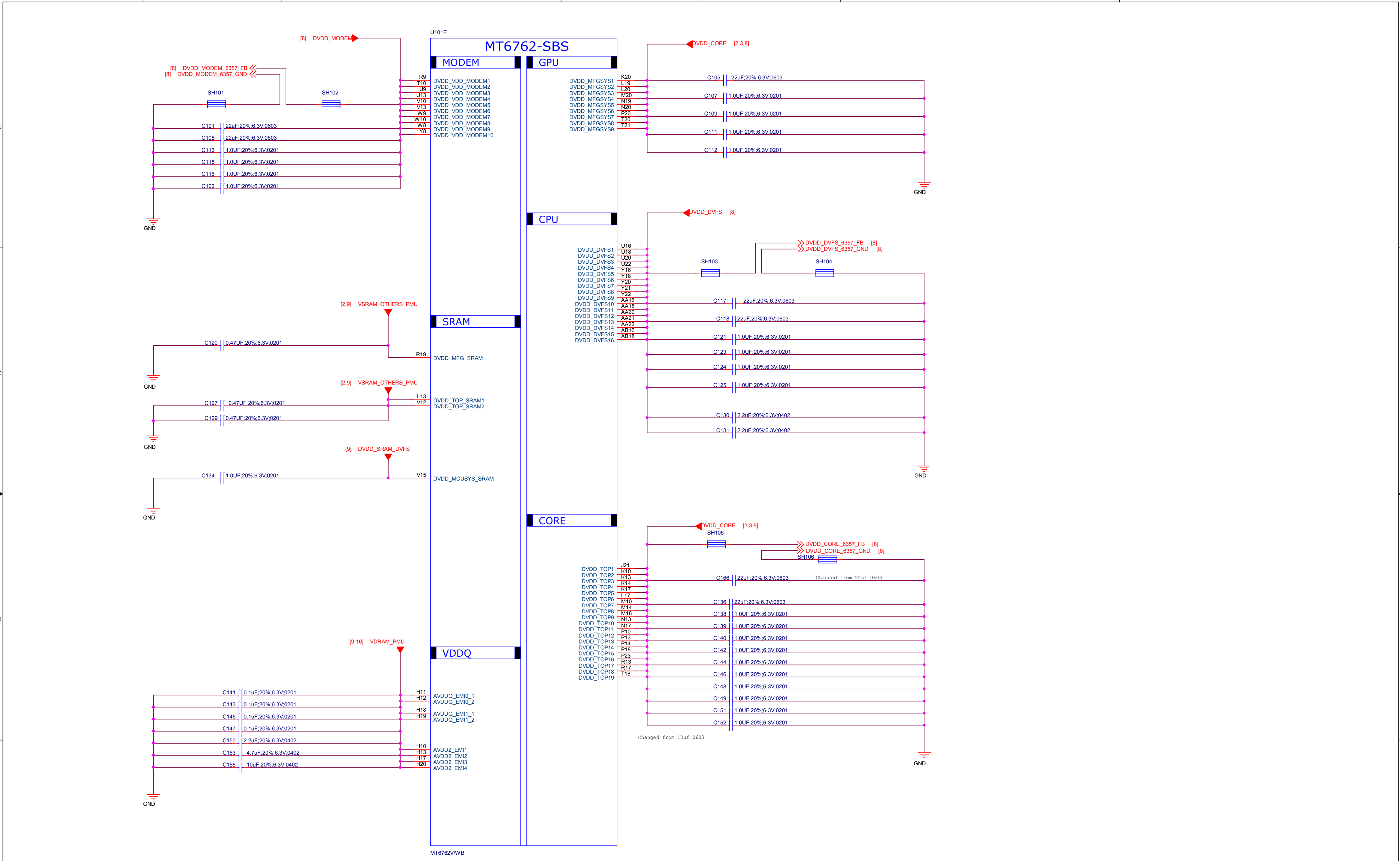
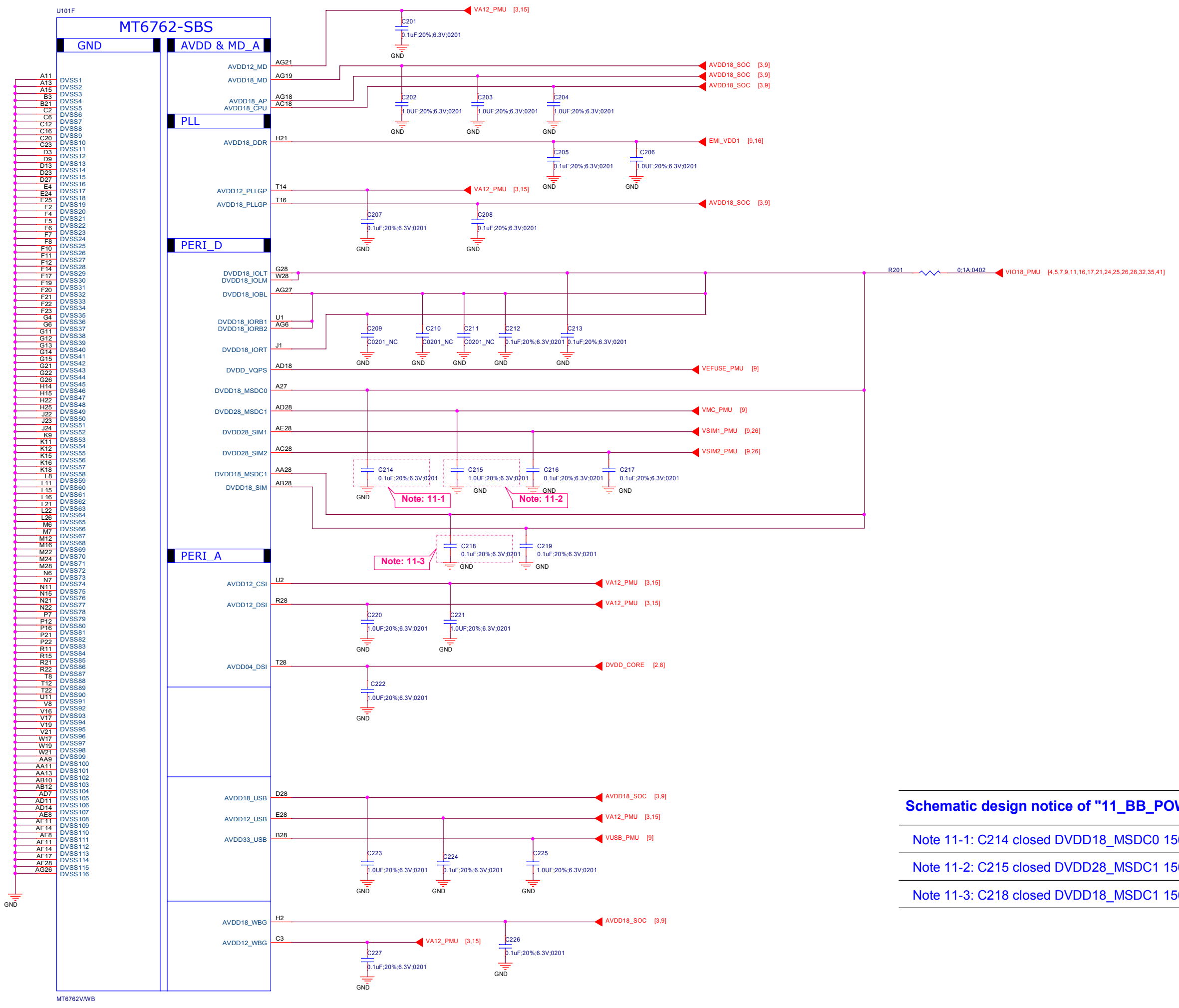
Revision History

REV	DATE	DESCRIPTION
A1	20181203	Refer to MTK DVT1.1 Reference design
A1	20181205	Change from K4 Sch 1. Change PA from 87318 to 87319 2.Change Charger IC from SM5414 to ETA6793 3.Change P-L sensor from STK3331 to Spring board STK3332 4. Delete LCM I2C switch 5. Delete USB switch 6.Delete KEY Pressed force download circuit 7.Change CAM I2C connection 8.Change BAT CONN

Title	00_History	REV: V10	
DOCUMENT NO.:	S96116-1-13_MB_20190409-1317	Size A2	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 1	of 53



Title	01_BB_ POWER_PDN		REV: V10
DOCUMENT NO.:	Design Name		Size A2
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 2	of 53



Schematic design notice of "11_BB_POWER_IO" page.

Note 11-1: C214 closed DVDD18_MSDC0 150mil

Note 11-2: C215 closed DVDD28_MSDC1 150mil

Note 11-3: C218 closed DVDD18_MSDC1 150mil

Title	02_BB_POWER_IO	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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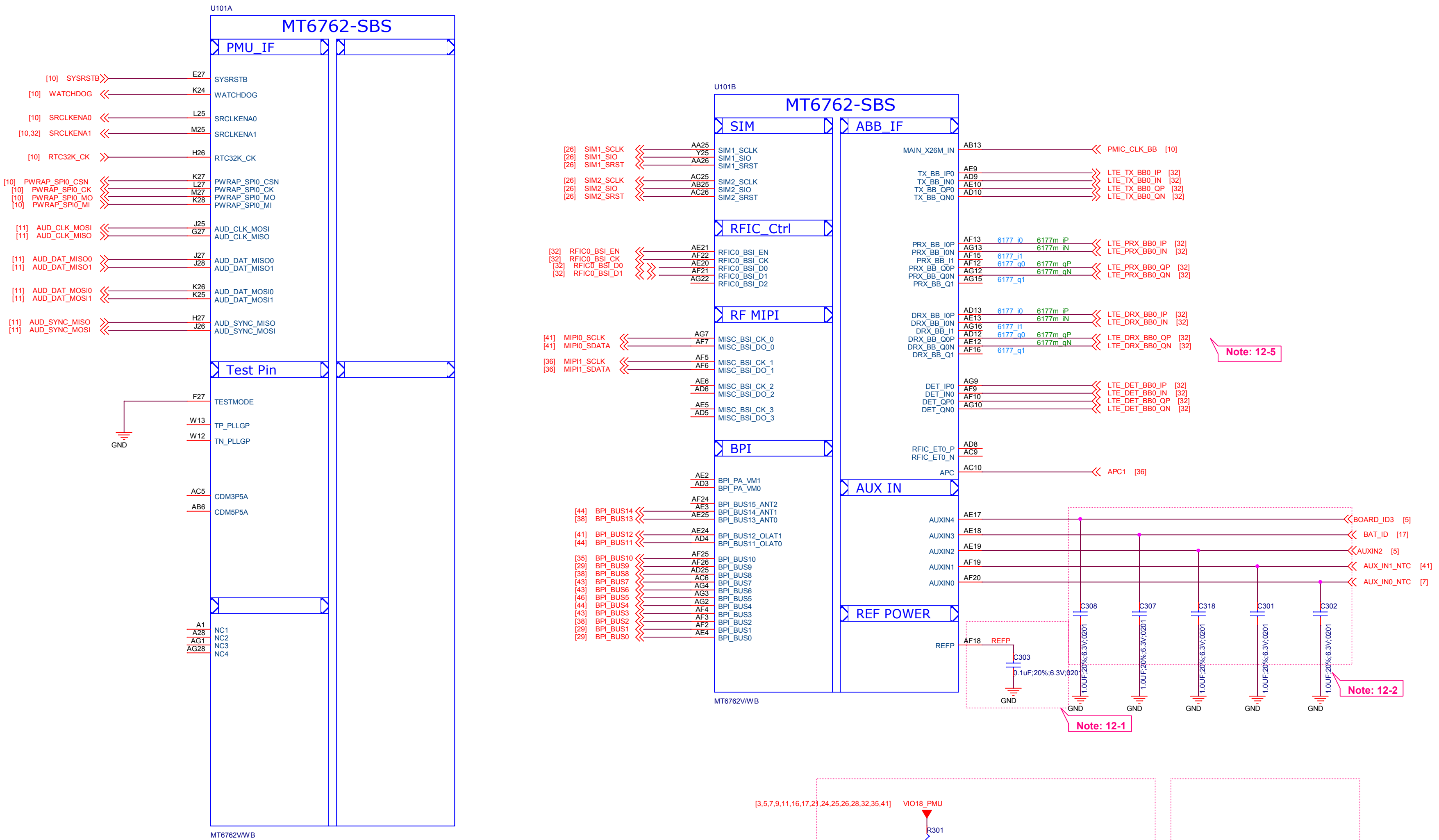

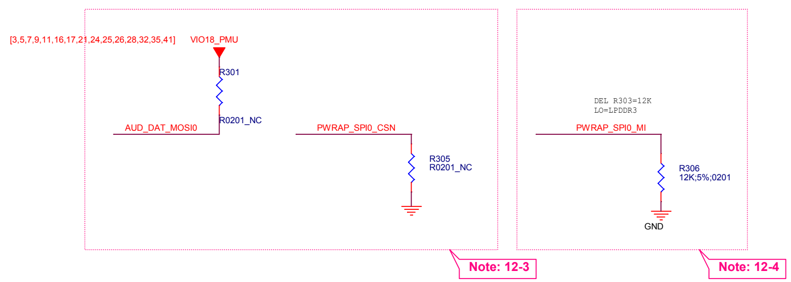

Schematic design notice of "12_BB_1" page.

Note 12-1: The de-coupling cap. for REFP (AF18 ball) have to be placed as close to BB as possible.

Note 12-2: To shunt a 1uF capacitor in the AUXIN ADC input to prevent noise coupling. It should be placed as close to BB as possible. Connect the unused AUX ADC input to GND.

Note 12-3: "PWRAP_SPI0_CSN" and "AUD_DAT_MOSI0" are bootstrap pin to select which interface will be the JTAG pin out.

PWRAP_SPI0_CSN	AUD_DAT_MOSI0	JTAG Function	
default=PU	default=PD	AP_JTAG	MD_JTAG
HI	LO	N/A	N/A
HI	HI (by ext. PU)	SPI0+EINT8	SPI1+SPI3
LO (by ext. PD)	LO	SPI0+EINT8	N/A
LO (by ext. PD)	HI (by ext. PU)	MSDC1	N/A

Note 12-4: PWRAP_SPI0_MI is DDR type feature in bootstrap

PWRAP_SPI0_MI	Booting interface	
default=PU	DDR	MSDC0 pin mux
LO (by ext. PD)	LPDDR3	follow LP3 Ref SCH.
HI	LPDDR4X	follow LP4X Ref SCH.

Note 12-5: Please set unused IQ pins in NC

Title	03_BB_1_RF&SIM_IF		REV: V10
DOCUMENT NO.:	Design Name		Size A2
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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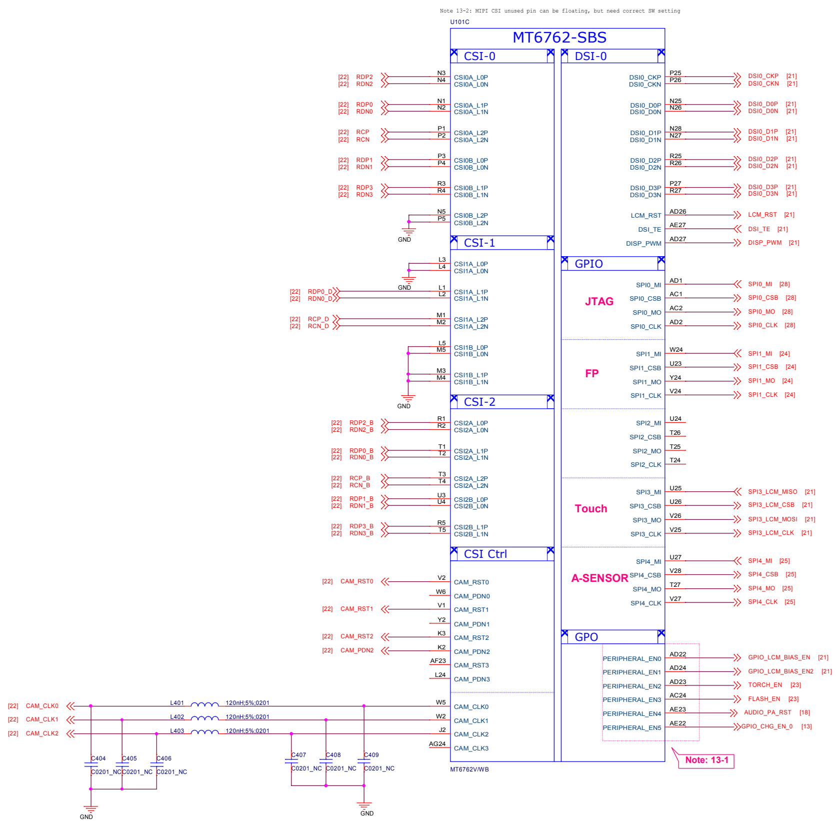
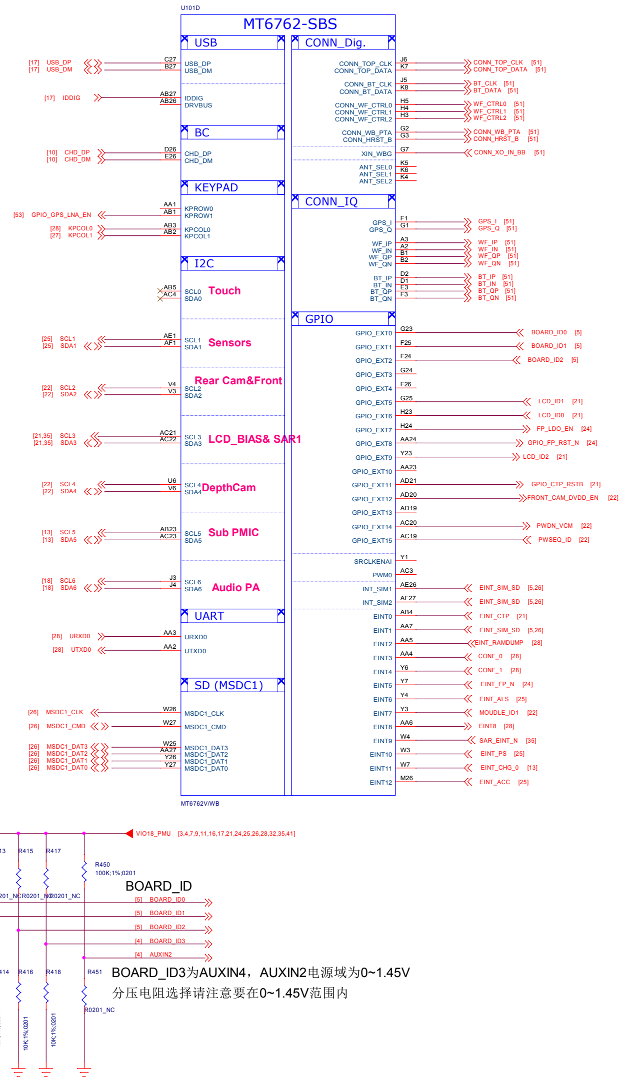
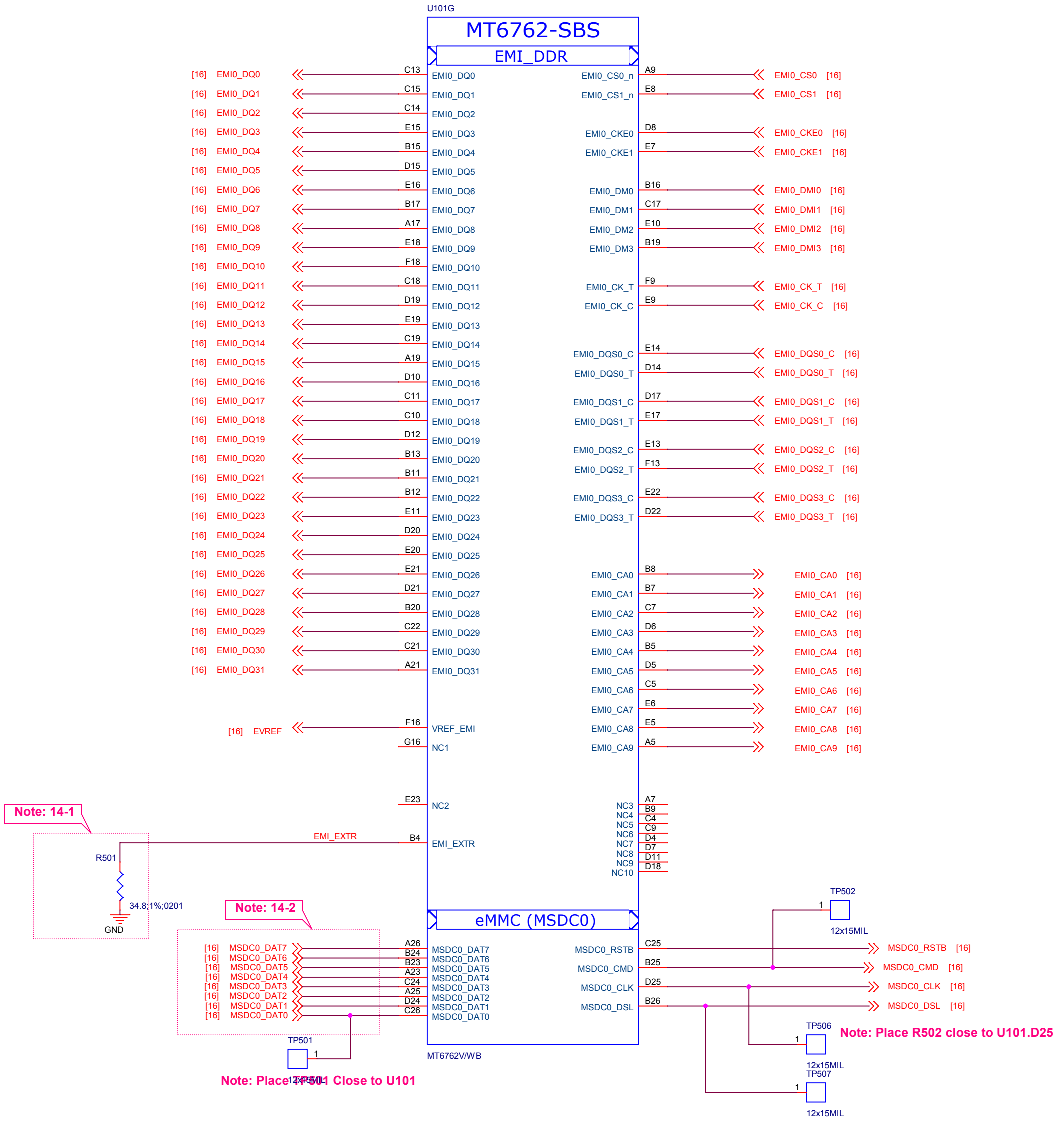




Schematic design notice of "13_BB_2" page.

Note 13-1: The enable pin of acoustic or optoelectronic devices (e.g. SPK AMP/Backlight/Charger OCP/OVP) suggest to use Peripheral_EN[0:5]
If use other GPIOs as enable pin, suggest to reserve 0201 NC to GND

Title	04_BB_2_MIPI&GPIO	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
Date: Monday, April 08, 2019	Sheet 5 of 53	



Schematic design notice of "14_BB_3" page.

Note 14-1: R501 please select 34.8 ohm (1%) resistor

Note 14-2: Please check eMCP LP3 and eMCP LP4X pin mux

Title	05_BB_3_EMMC		REV: V10
DOCUMENT NO.:	Design Name		Size A2
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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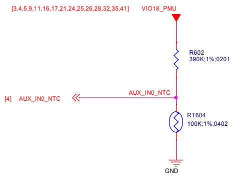
The pull up ressister need to be 1% accuracy
NTC=100K



Thermistor to sense CHARGER temperature

Huawei NOTE:1. NTC606 must keep a distance about 3~5 mm away from Charger IC and far from other heat sources 5~10 mm at least.

Thermistor to sense AP temperature

1. RT604must keep a distance about 6~8 mm away from AP and far from other heat sources 10 mm at least.
2. The distance is the shortest distance from package edge to edge.

Title	06_BB_AUXADC_Thermal	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI
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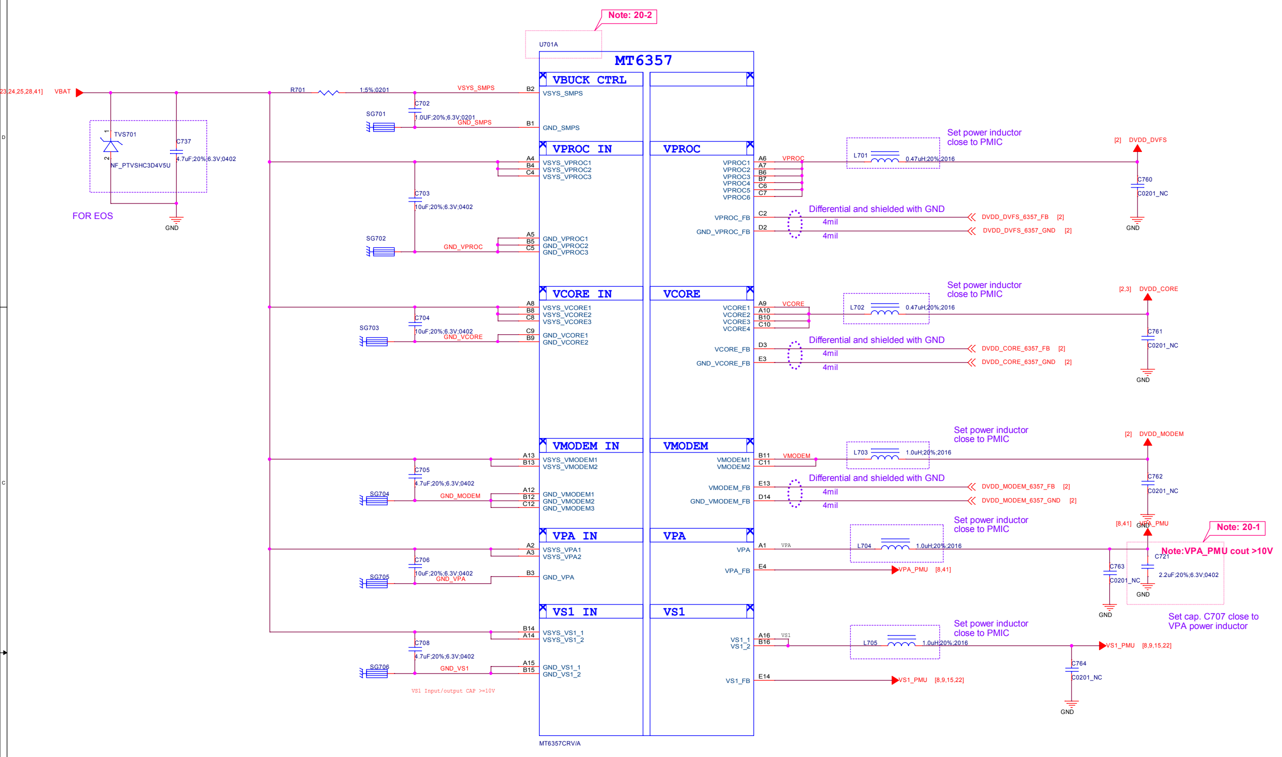

Schematic design notice of "20_POWER_MT6357_Buck"

Note 20-1: C707, please choose 0402 size

Note 20-2: For MT6765/62/61 platform, please only use MT6357CRV

Title	07_POWER_MT6357_Buck	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
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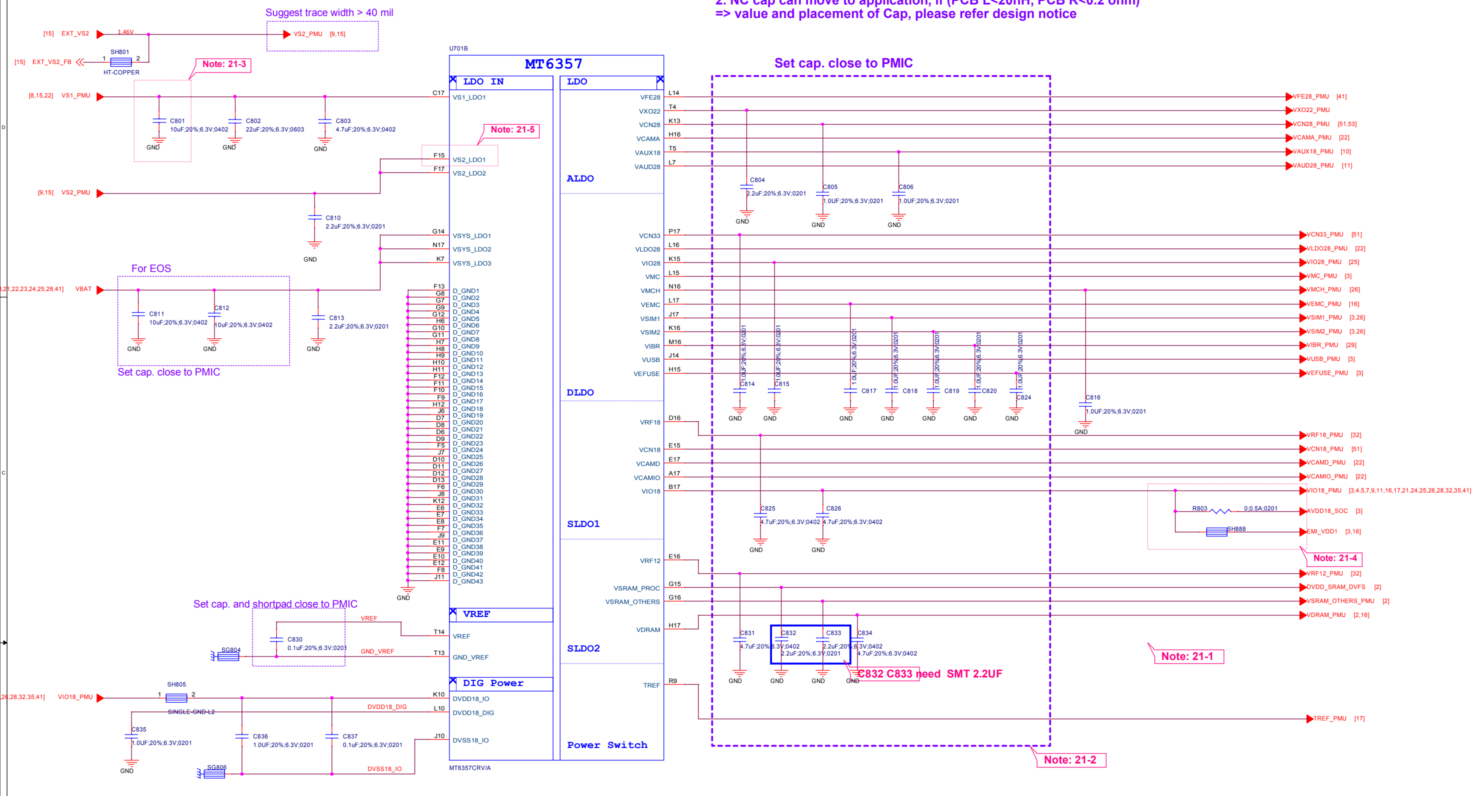
1. "Typical Cap" defined in design notice is the minimum cap. to LDO Cout.
2. NC cap can move to application, if (PCB L<20nH, PCB R<0.2 ohm)
=> value and placement of Cap, please refer design notice



Schematic design notice of "21_POWER_MT6357_LDO"

Note 21-1: If these power trace can meet LDO layout constraint, these CAP can be NC or removed. Please refer to MT6357 design notice.

Note 21-2: Output cap range please follow MT6357CRV LDO design notice

Note 21-3: Ext Buck BOM option

	Ext. buck option	
	w/ EXT VS2 Buck	w/o EXT VS2 Buck
C801	10uF	22uF

Note 21-4: Please set R803 and R803 close to C826, making star connection among VIO18_PMU, AVDD18_SOC, and EMI_VDD1 near to LDO cap. C826
Please also refer to MT6357 design notice for further detail design information

Note 21-5: Please connect VS2_LDO1(F15) to VS1_PMU if voltage applied to VCAMD(E17) >= 1.3 V

Title	08_POWER_MT6357_LDO		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI	
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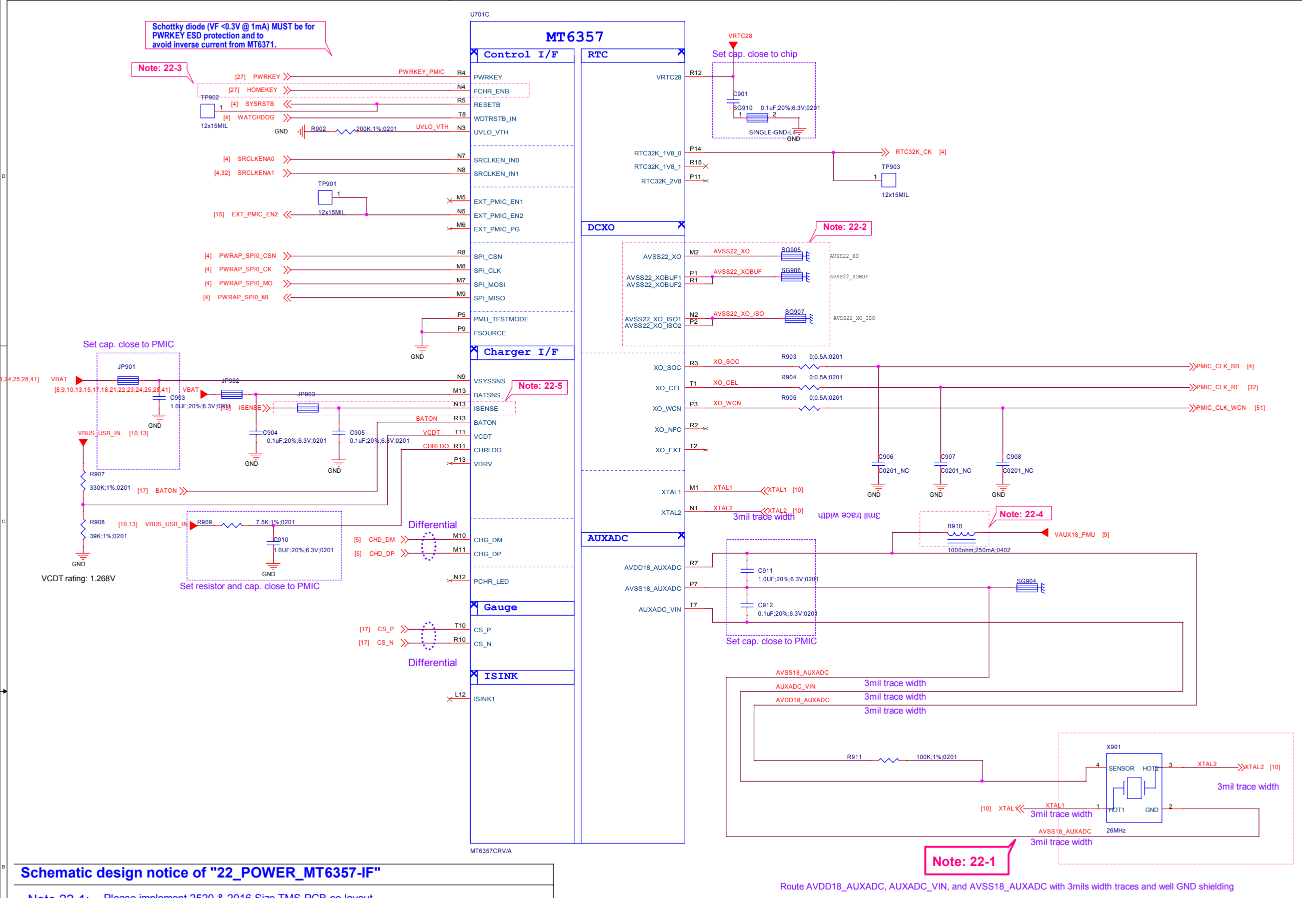

Schematic design notice of "22_POWER_MT6357-IF"

Note 22-1:	Please implement 2520 & 2016 Size TMS PCB co-layout. Please refer to MT6762_MT6357 Co-Clock Design Notice for co-layout guide
Note 22-2:	1. Please Connect P1 and R1 ball first and then to GND 2. Please Connect P2 and N2 ball first and then to GND 3. Please connect DCXO GND to main GND by independent L1-2 GND via.; DO NOT connect it through L1 GND
Note 22-3:	Let floating if disable HOMEKEY function
Note 22-4:	Please follow MT6762_MT6357 Co-Clock Design Notice for Layout guide of VAUX18, then R8101 can use 0 ohm to replace BEAD. Please route VAUX18_PMU with well-ground sheilding if using 0 ohm to replace BEAD for AVDD18_AUXADC
Note 22-5:	Please connect to battery connector

Title	09_POWER_MT6357_IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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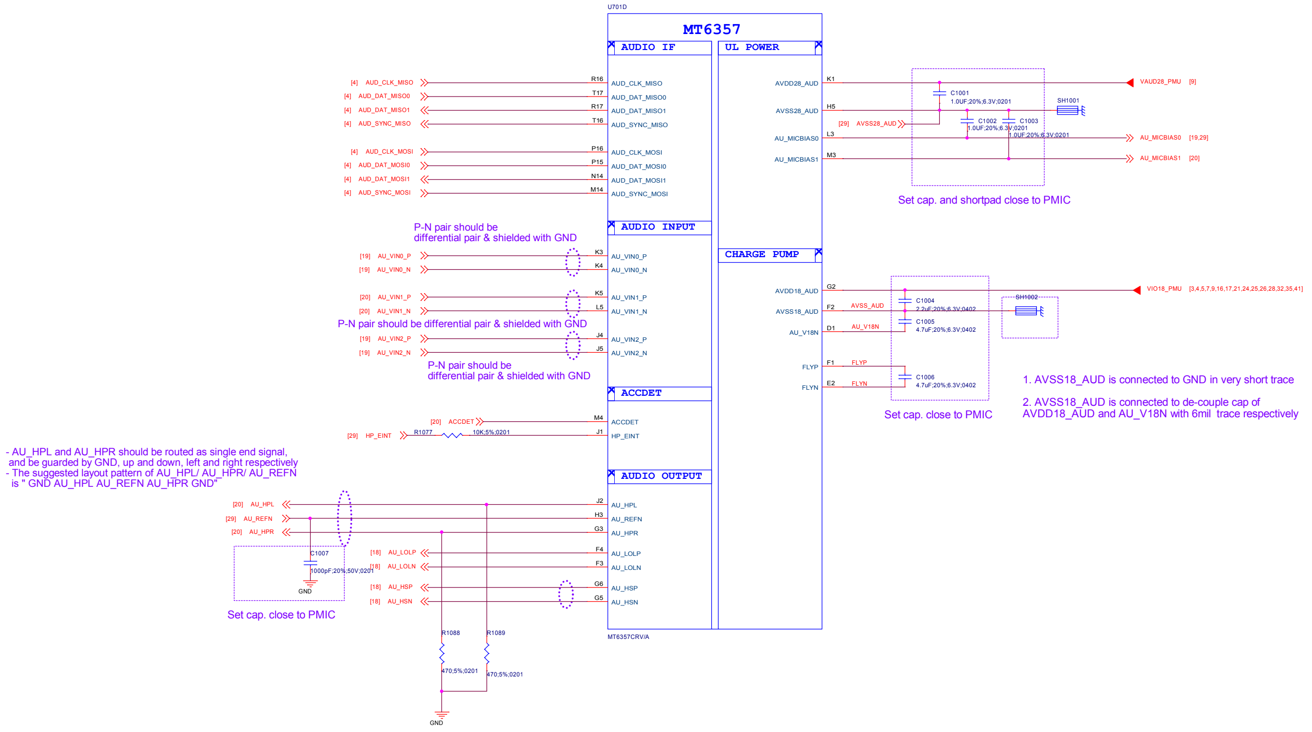

Title	10_POWER_MT6357_Audio	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI	
WINGTECH			
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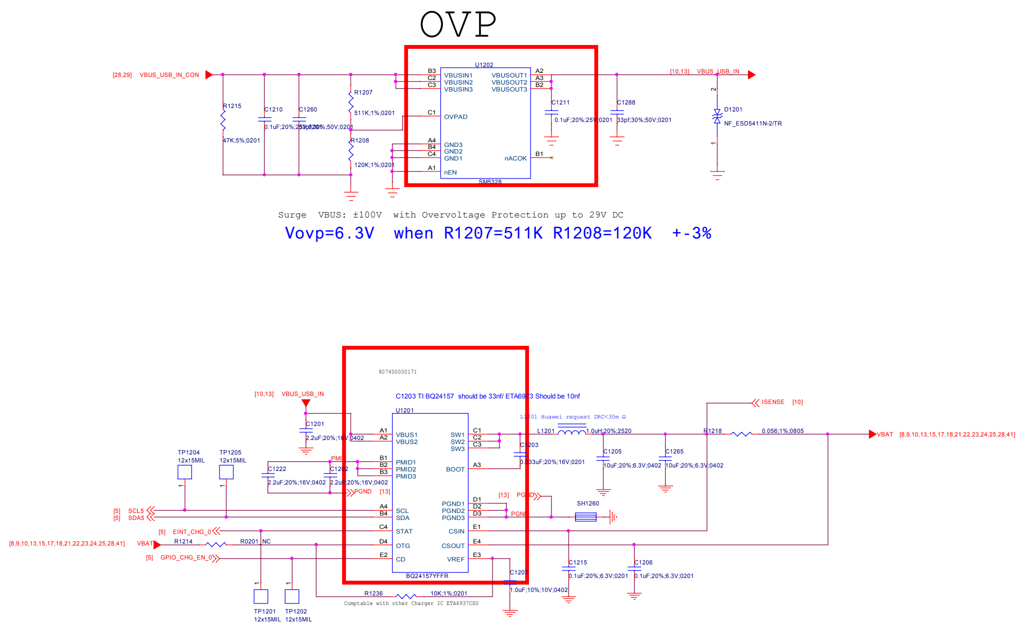
Title	11_POWER_SubPMIC-General	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	12 of 53



Schematic design notice of "26_POWER_SubPMIC-Charger + PP" page.

Note 26-1: For better ESD or surge performance we need choose suitable device for system protection. Please refer to [Surge device selection guide V2.0] provide by MTK.

Title	12_POWER_SubPMIC-Charger + PP
DOCUMENT NO.:	Design Name
Size	D
DEPARTMENT:	WINGTECH-SH
DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019
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Title	13_POWER_SubPMIC-HV powers	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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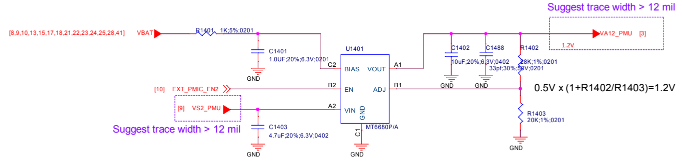
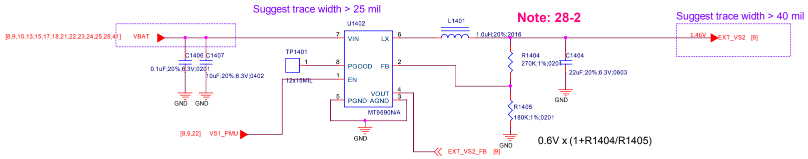
LDO for VA12

Note: 28-1



Ext. Bulk for VS2

Note: 28-2



NOTE: Do not use VCAMD 1.3/1.5/1.8 when VS2 BUCK applied

Schematic design notice of "28_POWER_ThirdParty-Power"

Note 28-1:	VA12 Layout placement please close to AP
Note 28-2:	VS2 Buck Layout placement please close to PMIC MT6357
Note 28-3:	VCN33 LDO Layout placement please close to MT6631

Title	14_POWER_ThirdParty_Powers	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI	
WINGTECH			
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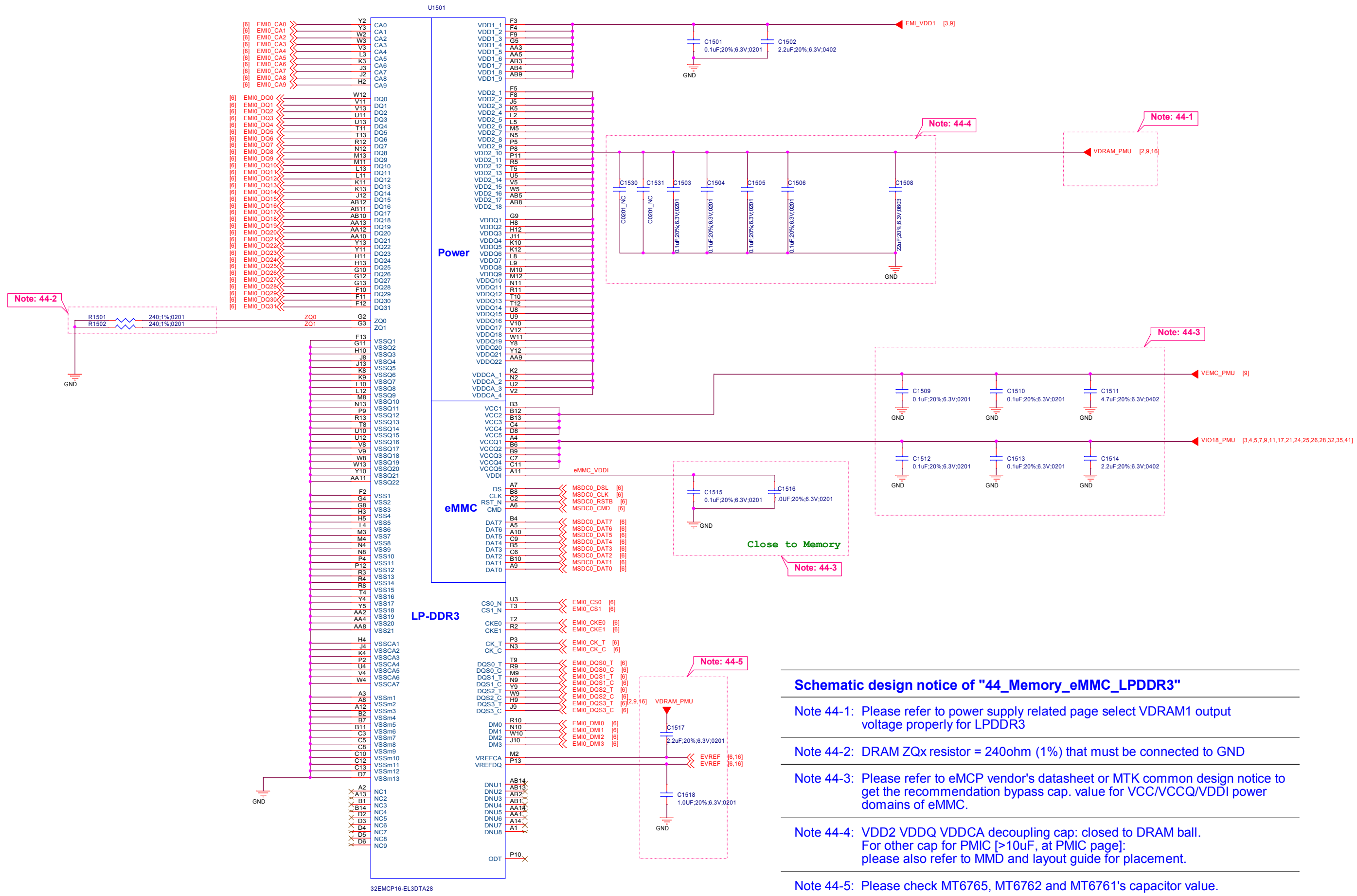
LPDDR3



Schematic design notice of "44_Memory_eMMC_LPDDR3"

Note 44-1: Please refer to power supply related page select VDRAM1 output voltage properly for LPDDR3

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to GND

Note 44-3: Please refer to eMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDDI power domains of eMMC.

Note 44-4: VDD2 VDDQ VDDCA decoupling cap: closed to DRAM ball. For other cap for PMIC [>10uF, at PMIC page]: please also refer to MMD and layout guide for placement.

Note 44-5: Please check MT6765, MT6762 and MT6761's capacitor value.

Project	C1517	C1518
MT6765	2.2uF	1uF
MT6762	2.2uF	1uF
MT6761	0.1uF	0.1uF

Title: 15_Memory_eMMC_LPDDR3		REV: V10
DOCUMENT NO.: Design Name		Size: D
DEPARTMENT: WINGTECH-SH		DESIGNER: YANGSHAOYUN
WINGTECH		
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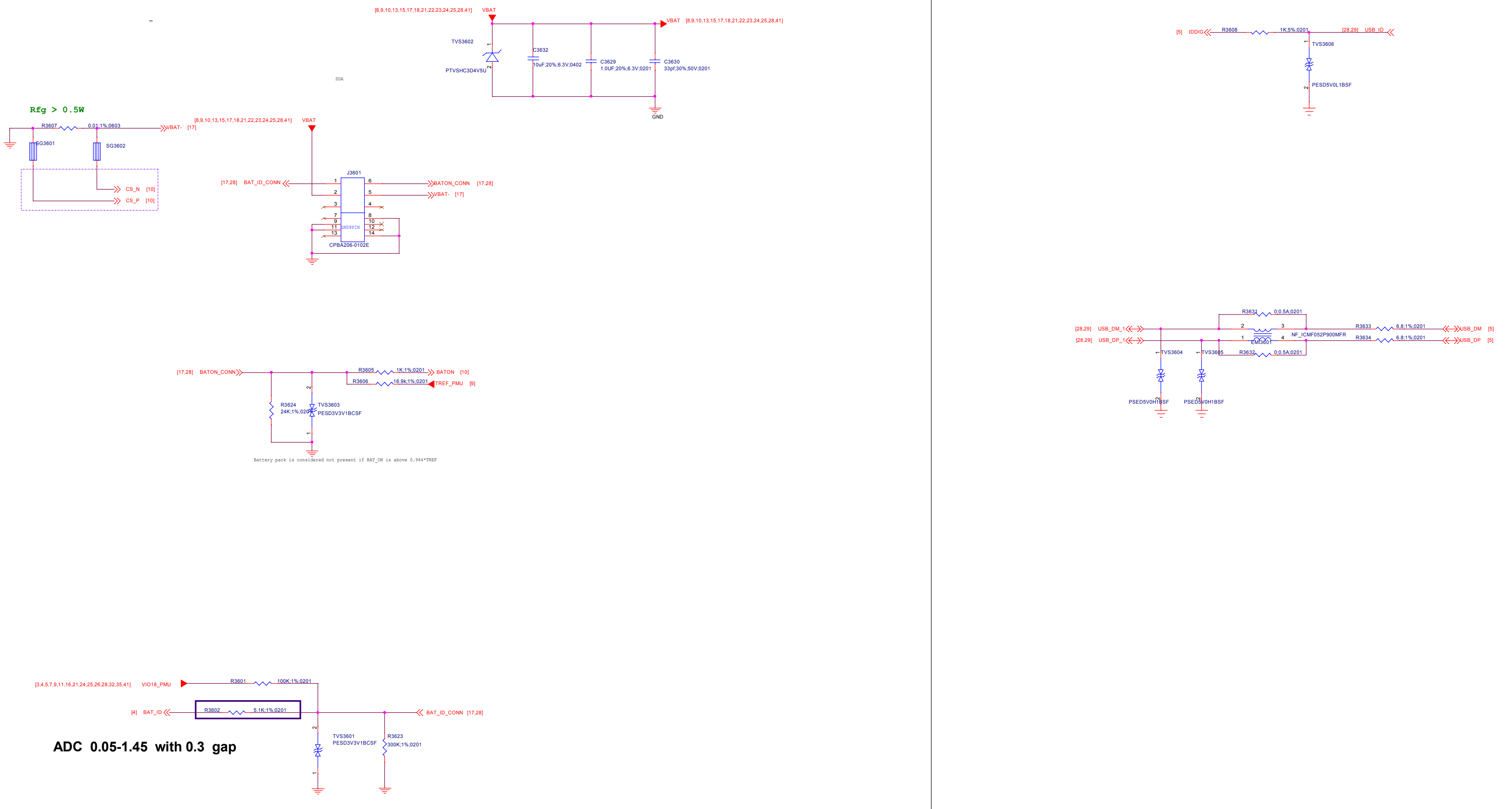
BATTERY CONNECTOR



Title	36_Battery/USB IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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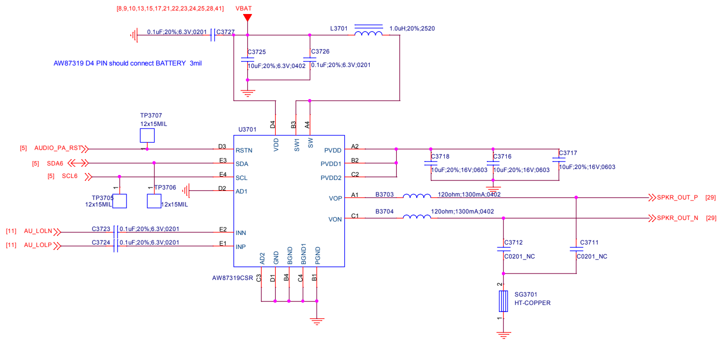
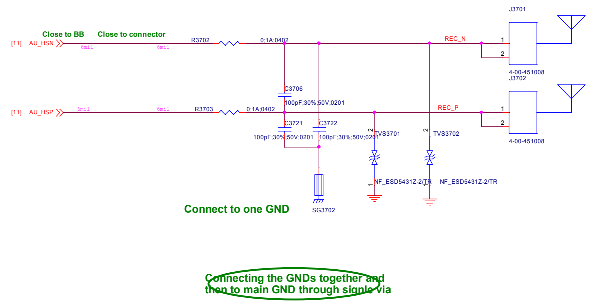
SPK



Receiver



Title	37_SPK/Receiver/Vibrator	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH	DESIGNER: LIUFENGLEI	
WINGTECH		
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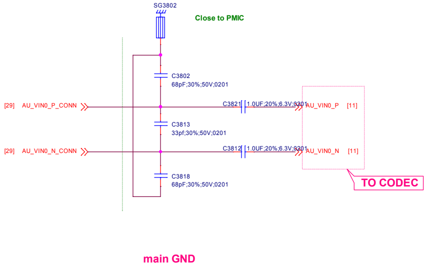
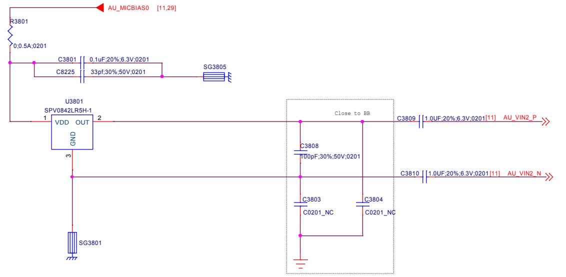
MAIN MIC



TOP MIC



Title	38_MIC	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI	
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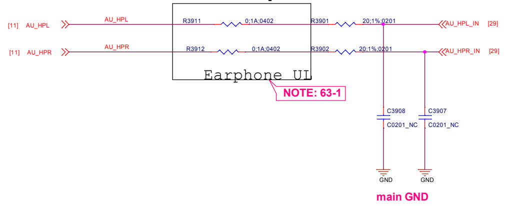
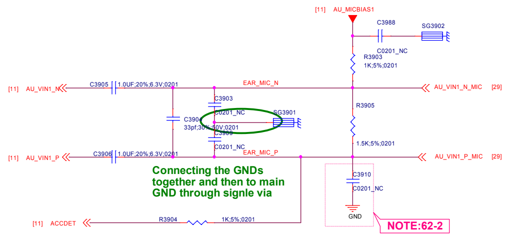
Earphone





NOTE:62-1 R3911 R3912, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Title	39_Earphone	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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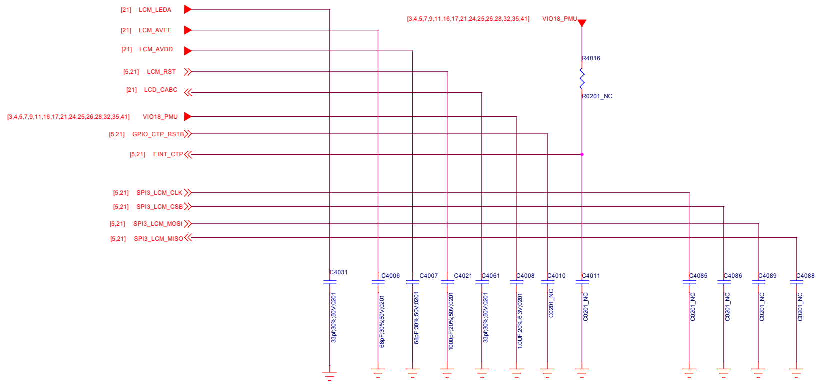
Note 62-1: Part # of BEAD6202, BEAD6203, BEAD6204 and BEAD6205 needs changed to "BLM18BD102SN1" for high THD performance (-90dB) but this BOM change will results in FM RSSI 10dB degraded .

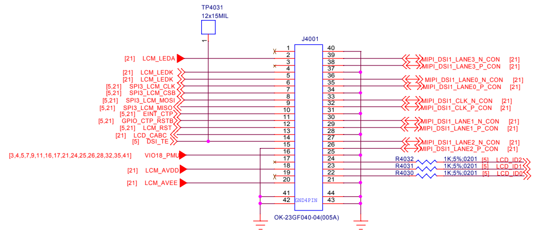
Note 62-2: Reserved Cap for CS/RS test, please double check multi-key function when used

Note 62-3:

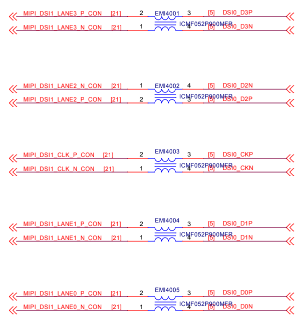
	Earphone Jack @ Main board	Earphone Jack @ Sub board
R6513	0 ohm	100nF

Note 62-4: Please Select ACC Mode for Operator Project to Pass Electrical MOS Test;

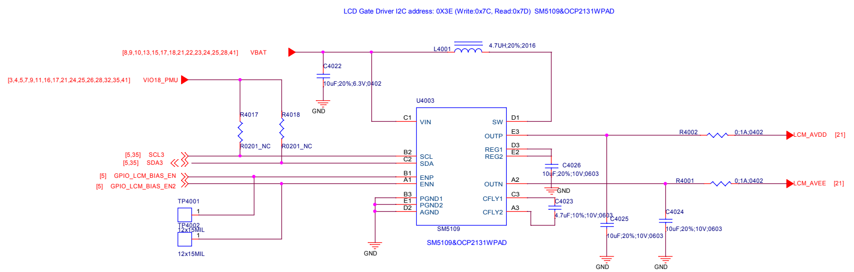




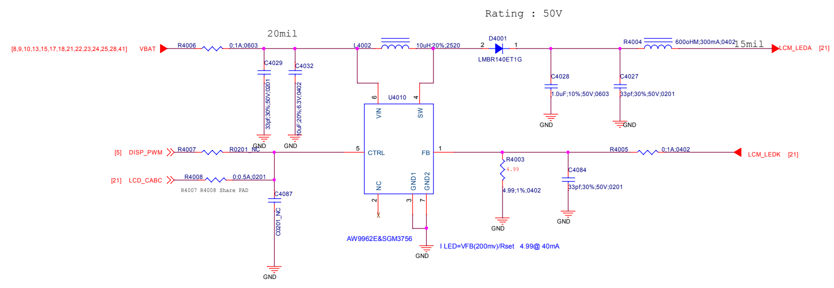
Common Mode Filter



LCD Gate Drive



LCD Backlight LED Driver



Title	40_LCD/CTP IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
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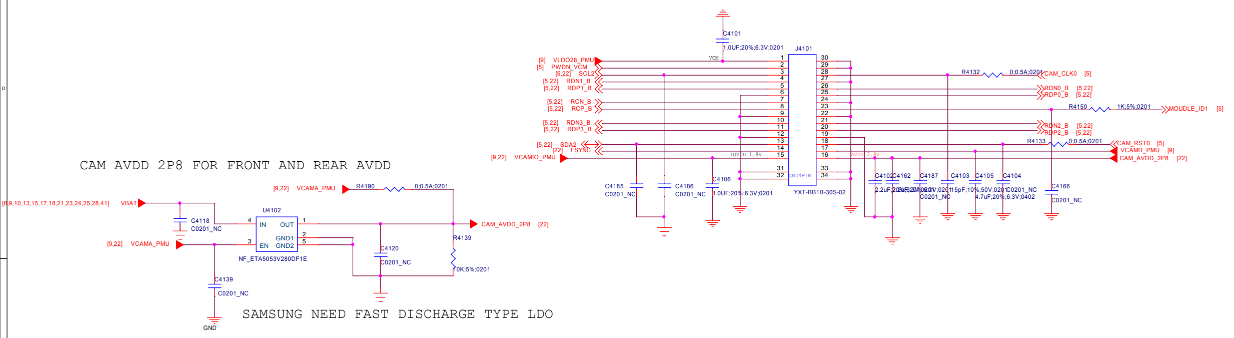
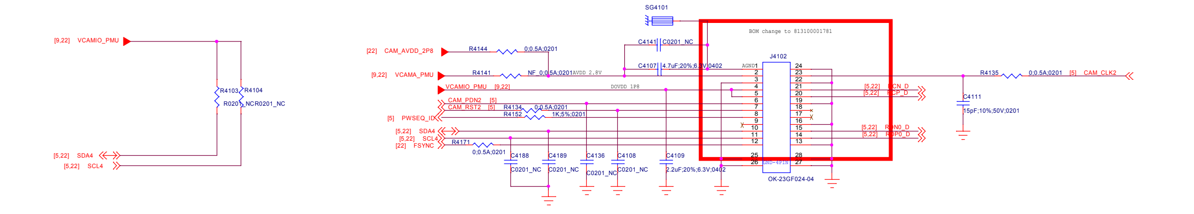
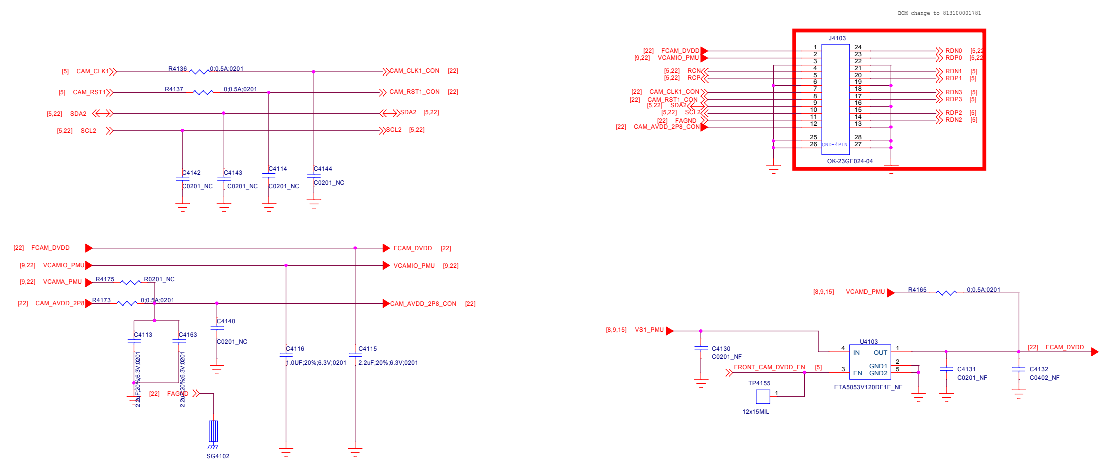
Main Camera 13M

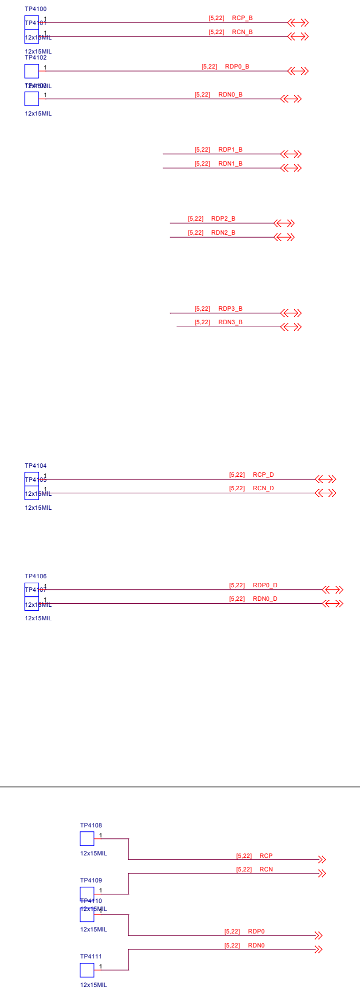


Depth Camera 2M



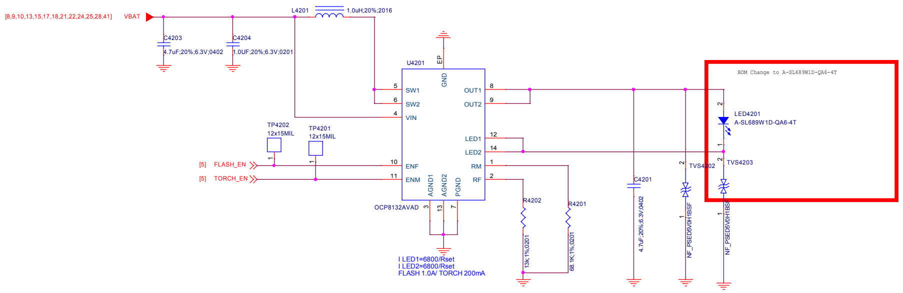
Front Camera 8M



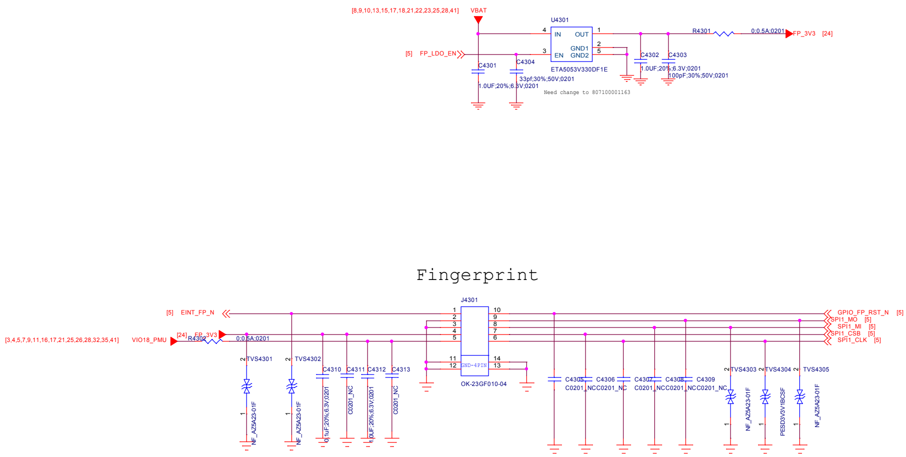


Title	41_Camera IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Tuesday, April 09, 2019	Sheet 22 of 53	

WINGTECH

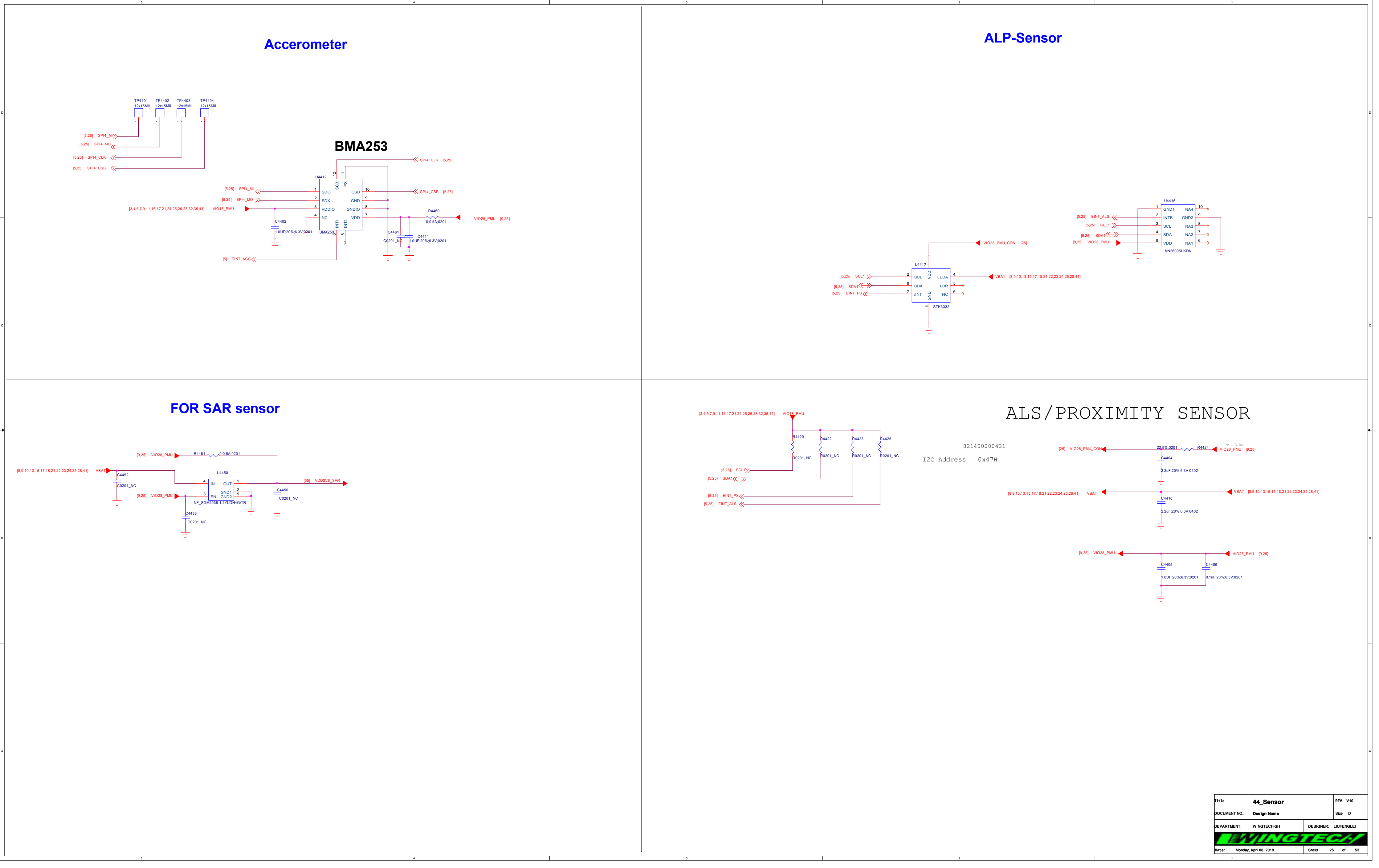


Title	42_Flash/RGB	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 23	of 53



Fingerprint

Title	43.Fingerprint	REV: V10	
DOCUMENT NO.:	S96116-1-13_MB_20190409-1317	Size D	
DEPARTMENT:	Wingtech-SZ	DESIGNER:	DESIGNER
Date:	Monday, April 08, 2019	Sheet 24 of 53	


Accerometer
BMA253
U4412
SPI4_MI
SPI4_MO
SPI4_CLK
SPI4_CSB
VIO18_PMU
VIO28_PMU
EINT_ACC
TP4401
TP4402
TP4403
TP4404
12x15MIL
C4402
C4461
C4411
R4460
0:0.5A;0201
C0201_NC
1.0UF;20%;6.3V;0201
ALP-Sensor
U4416
MN26005UKDN
EINT_ALS
SCL1
SDA1
U4417
STK3332
VIO28_PMU_CON
VBAT
EINT_PS
FOR SAR sensor
U4450
NF_SGM2036-1.2YUDH4G/TR
R4461
C4450
C4452
C4453
VDD2V8_SAR
ALS/PROXIMITY SENSOR
821400000421
I2C Address 0x47H
R4420
R4422
R4423
R4425
R0201_NC
R4424
C4404
C4410
C4405
C4406
2.2uF;20%;6.3V;0402
0.1uF;20%;6.3V;0201
Title 44_Sensor
REV: V10
DOCUMENT NO.: Design Name
Size D
DEPARTMENT: WINGTECH-SH
DESIGNER: LIUFENGLEI
Date: Monday, April 08, 2019
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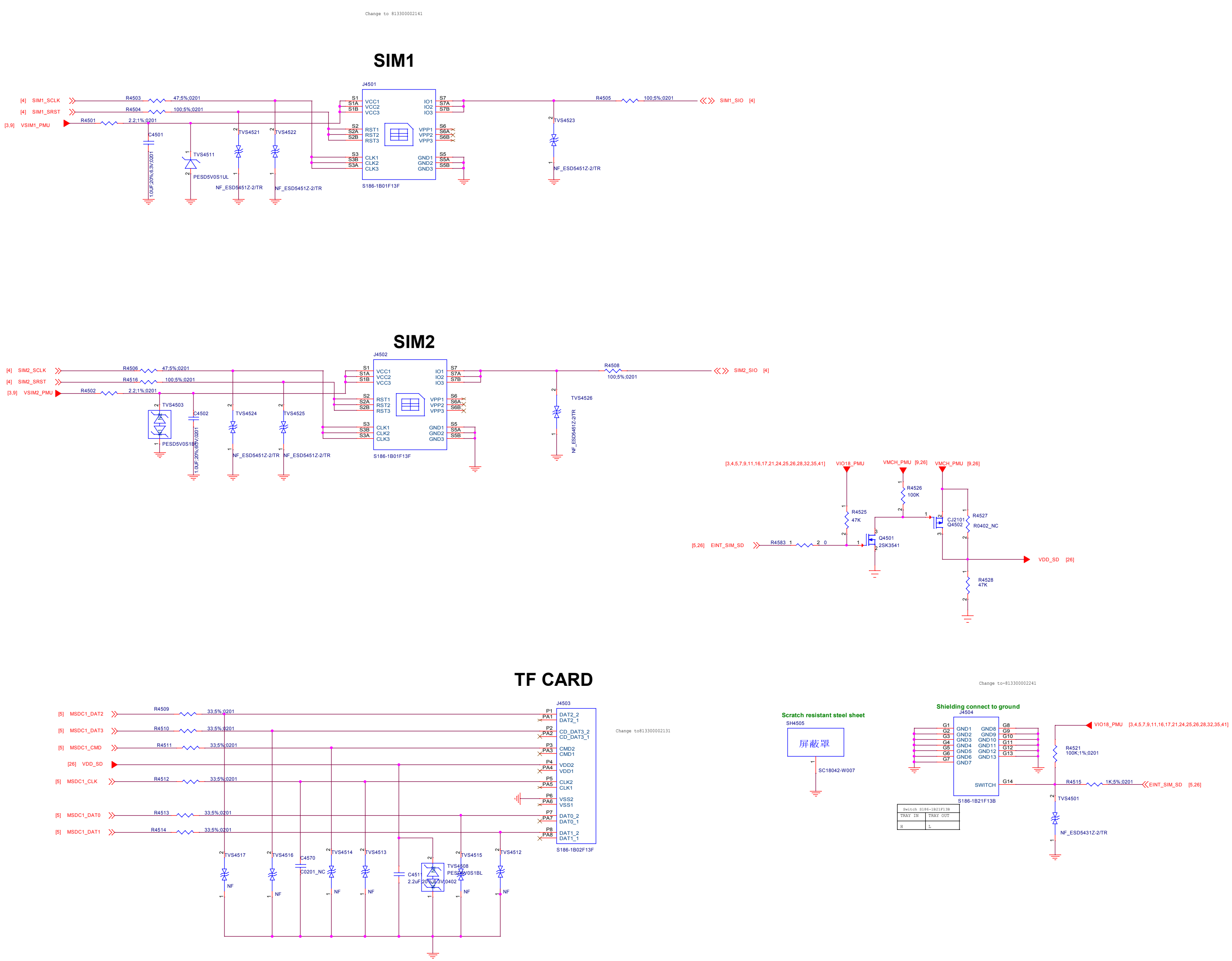
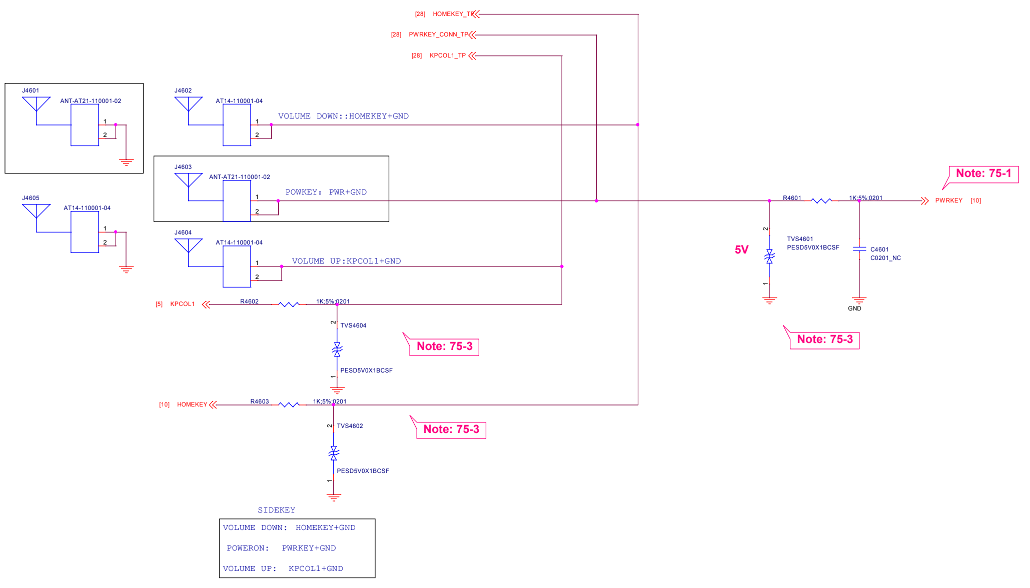


Title	45_SIM/TF IF	REV:	V10
DOCUMENT NO.:	Design Name	Size	D
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet	26 of 53

WINGTECH

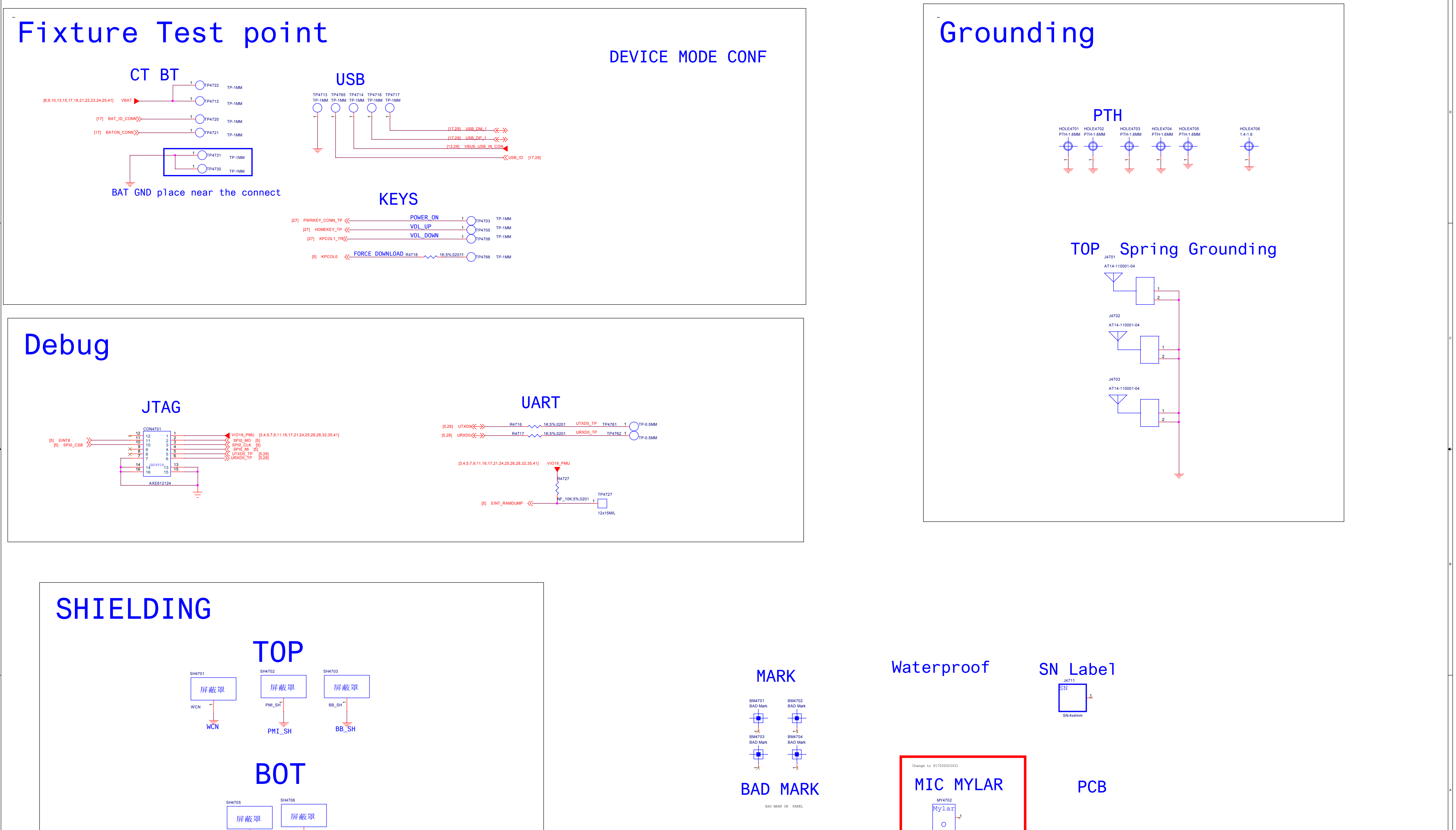
Power Key / Key Pad

DO NOT put pull-up resistor on PWRKEY

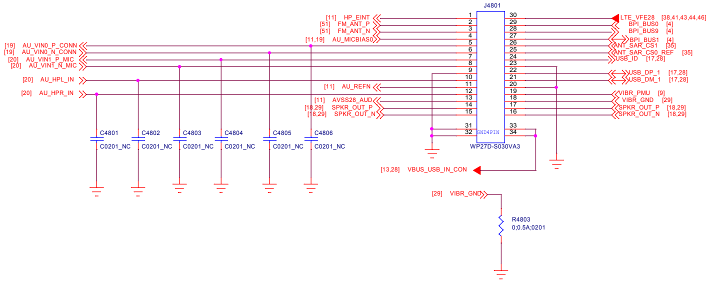


Note 75-1:DO NOT put pull-up resistor on PWRKEY
Note 75-2: Volume Up:HOME KEY+GND
Volume Down:KPCOL1+GND

Title	46_Sidekey	REV: V10	
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Thursday, April 11, 2019	Sheet 27 of 53	

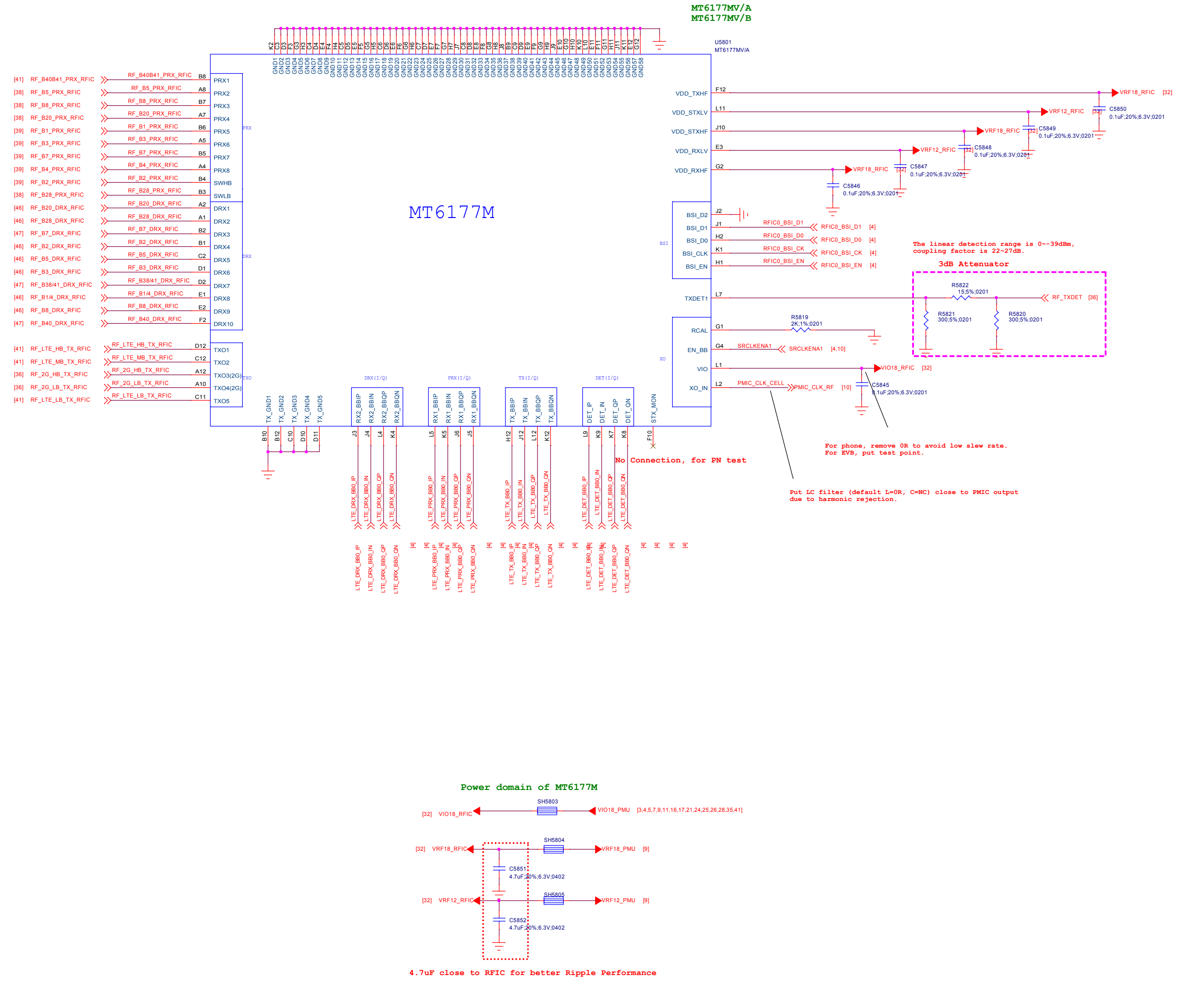


Title	47_Testpoint/Shielding/GND	REV: V10
DOCUMENT NO.:	Design Name	Size D
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI
Date: Monday, April 08, 2019		Sheet 28 of 53



Title	48_Sub PCB IF		REV: V10
DOCUMENT NO.:	Design Name		Size D
DEPARTMENT:	WINGTECH-SH	DESIGNER: LIUFENGLEI	
Date:	Monday, April 08, 2019	Sheet 29 of 53	

Title	<Title>	
Size A1	Document Number <Doc>	Rev 1.0
Date: Monday, April 08, 2019	Sheet 31 of 53	

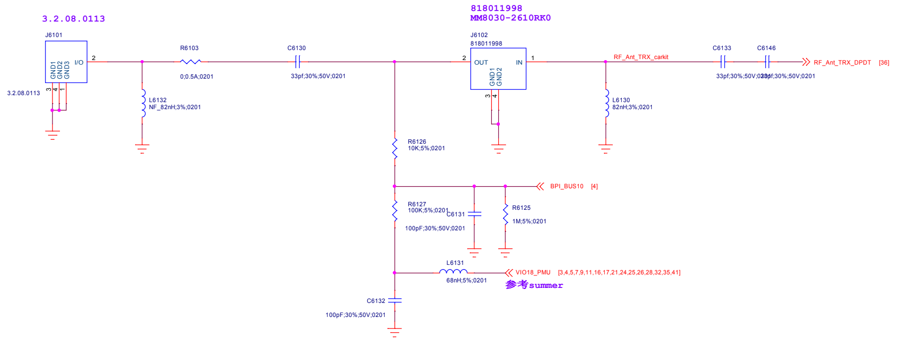


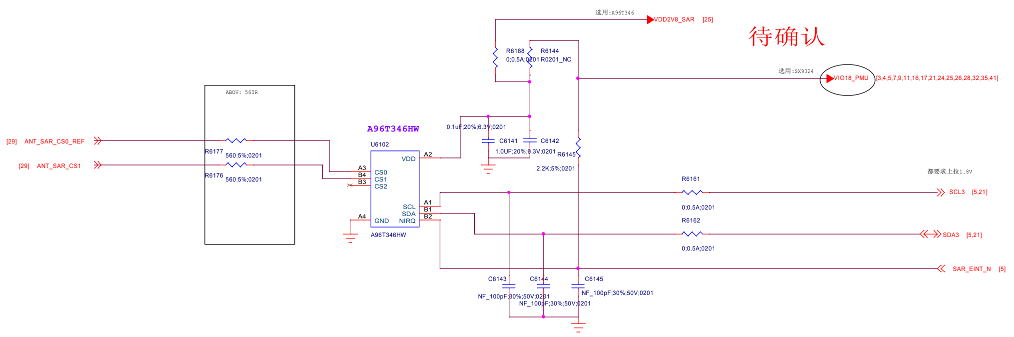
Title	Page Name = 58_Transceiver1
REV:	V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190408-1317
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ
DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019
Sheet	32 of 53

WINGTECH

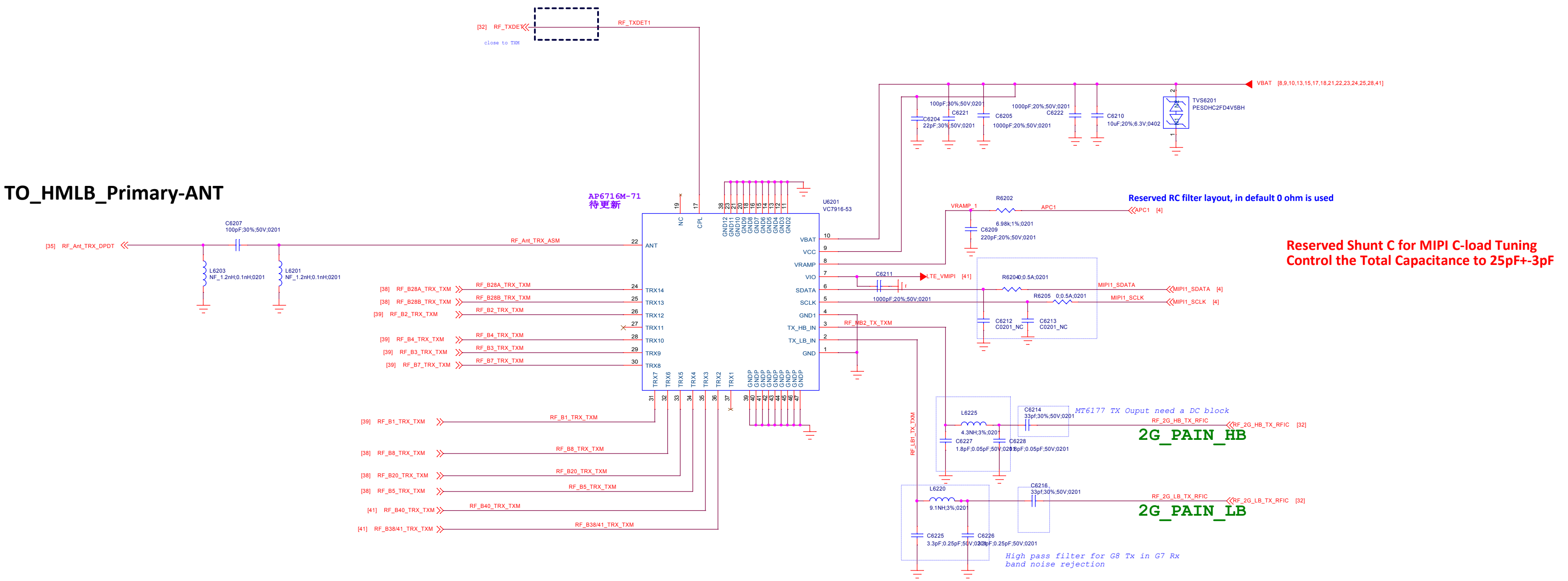
Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 33	of 53

Title	Reserved	REV: V10
DOCUMENT NO.:	Design Name	Size A1
DEPARTMENT: WINGTECH-SH		DESIGNER: LIUFENGLEI
Date: Monday, April 08, 2019		Sheet 34 of 53





Title	REV: V10
DOCUMENT NO.:Design Name = S96116-1-13_MB_20190409-1317	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
WINGTECH	
Date: Page Modify Date = Monday, April 08, 2019	Sheet 35 of 53



Title	Page Name = 62_Primary_ASM
REV:	V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1317
DEPARTMENT:	DEPARTMENT = WINGTECHSZ
DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019
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5 4 3 2 1

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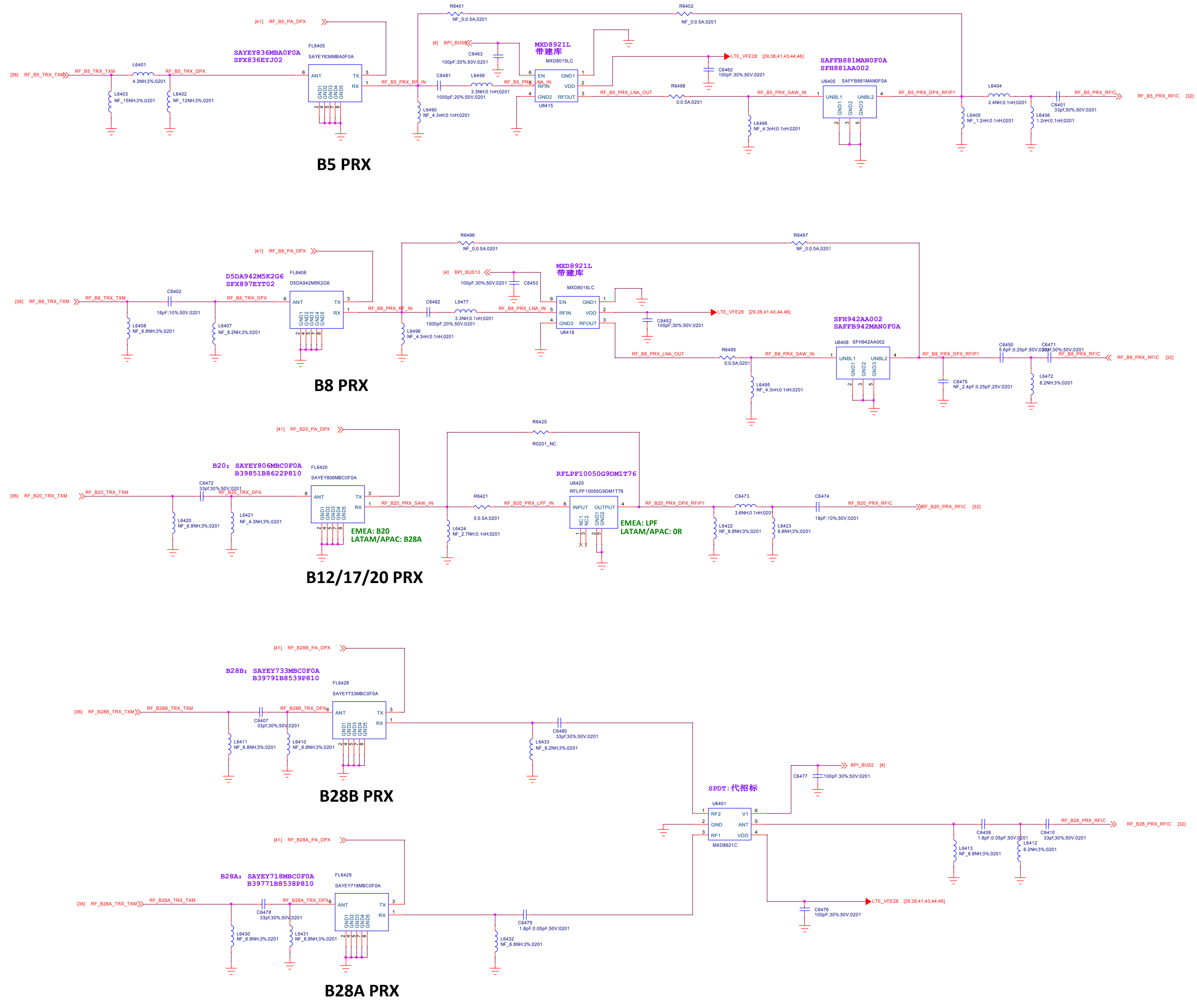
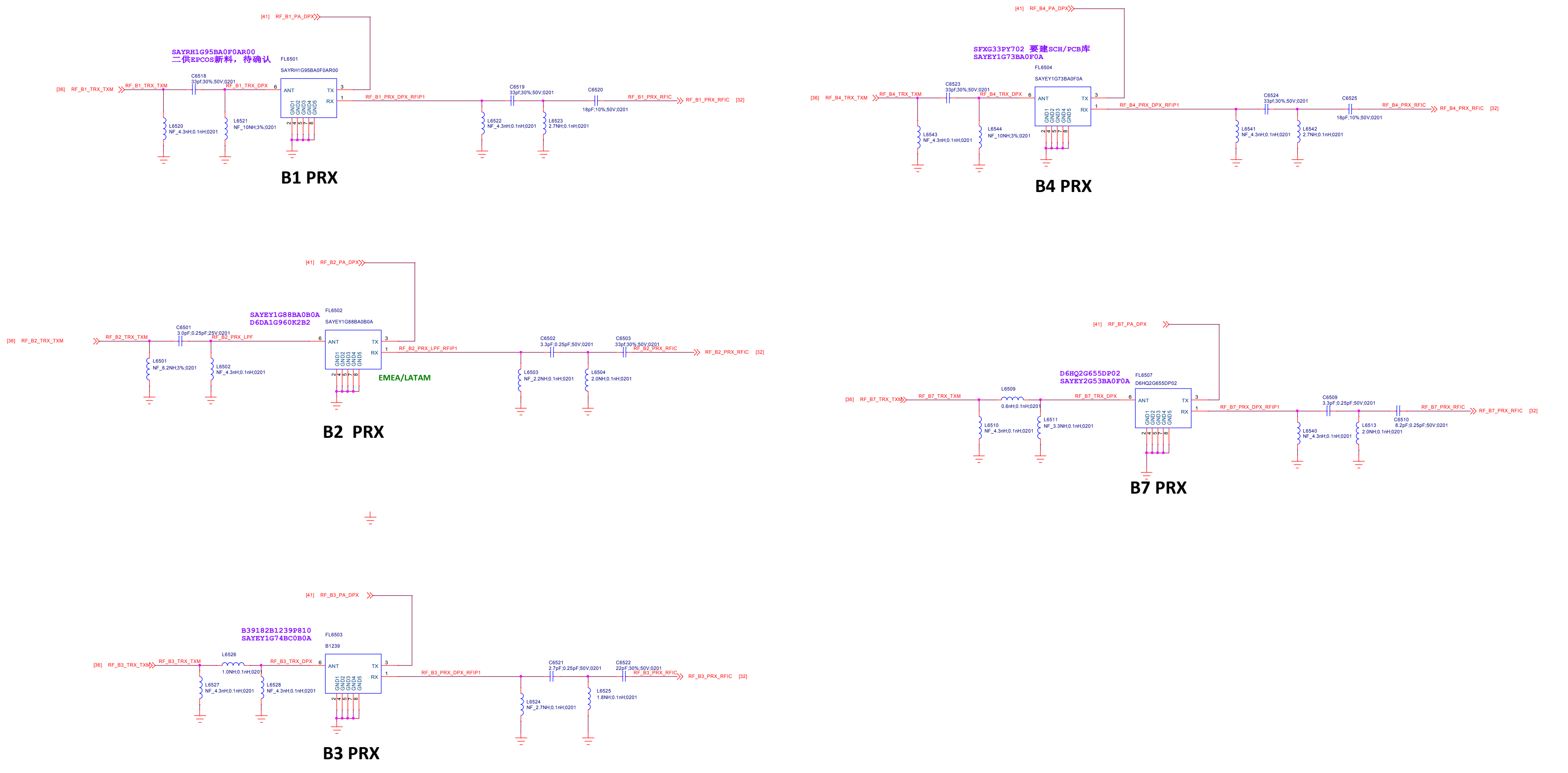
Title	<Title>	
Size A1	Document Number <Doc>	Rev 1.0
Date: Monday, April 08, 2019	Sheet 37 of 53	

5 4 3 2 1



Title	Page Name = 64_TRX_LB — REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190408-1317
DEPARTMENT:	DEPARTMENT = WINGTECHSZ — DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019 — Sheet 38 of 53



Title	Page Name = 65_TRX_MHB	REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-0317	
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI
WINGTECH		
Date:	Page Modify Date = Monday, April 08, 2019	Sheet 39 of 53

5 4 3 2 1

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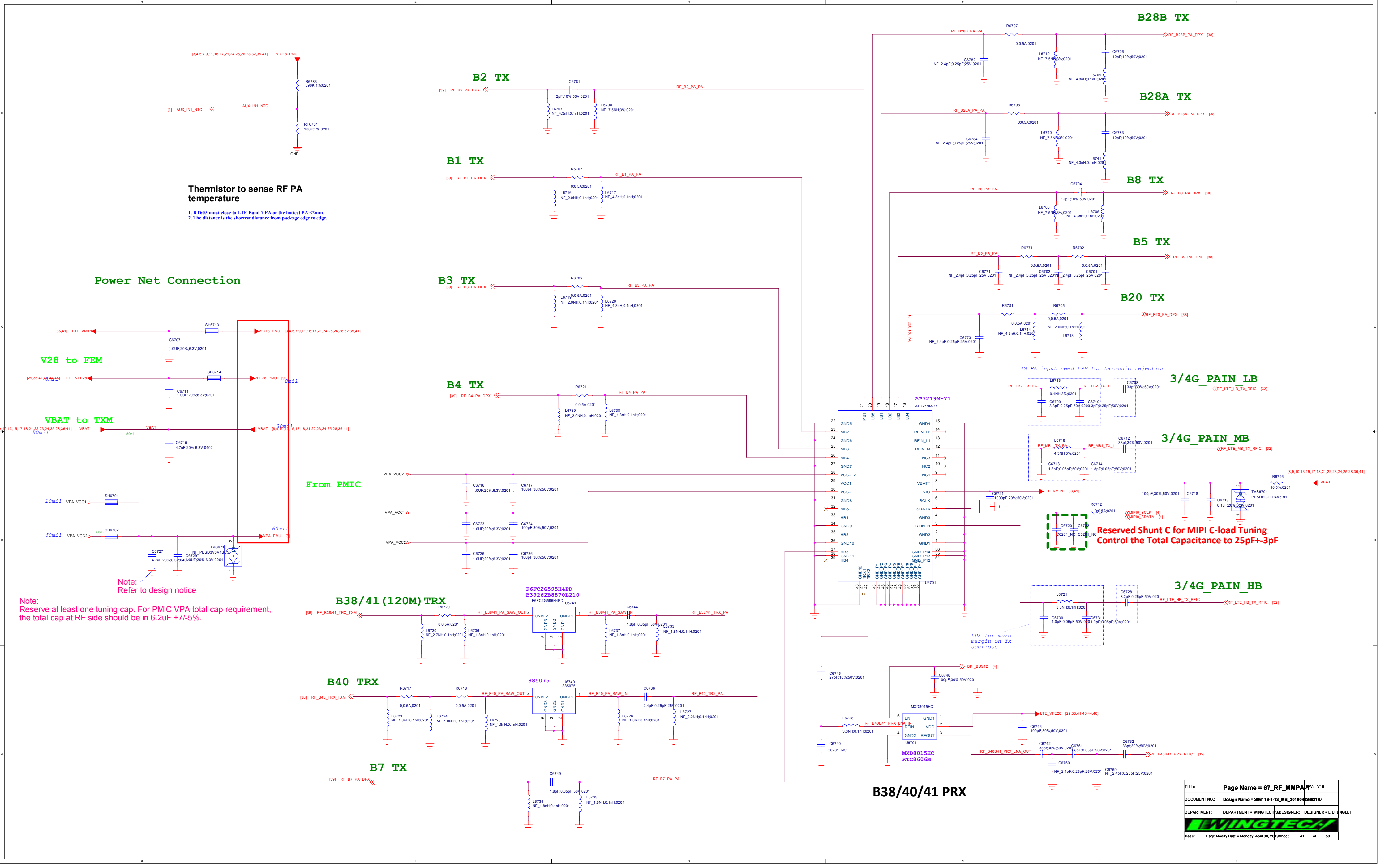
Title <Title>		
Size A1	Document Number <Doc>	Rev 1.0
Date: Monday, April 08, 2019	Sheet 40 of 53	

5 4 3 2 1


B28B TX
B28A TX
B8 TX
B5 TX
B20 TX
B2 TX
B1 TX
B3 TX
B4 TX
B38/41(120M)TRX
B40 TRX
B7 TX
Thermistor to sense RF PA temperature
1. RT603 must close to LTE Band 7 PA or the hottest PA <2mm.
2. The distance is the shortest distance from package edge to edge.
Power Net Connection
V28 to FEM
VBAT to TXM
From PMIC
AP7219M-71
4G PA input need LPF for harmonic rejection
3/4G_PAIN_LB
3/4G_PAIN_MB
3/4G_PAIN_HB
Reserved Shunt C for MIPI C-load Tuning
Control the Total Capacitance to 25pF+-3pF
LPF for more margin on Tx spurious
MXD8015HC
RTC8606M
B38/40/41 PRX
F6FC2G595H4PD
B39262B8870L210
885075
Note:
Refer to design notice
Note:
Reserve at least one tuning cap. For PMIC VPA total cap requirement, the total cap at RF side should be in 6.2uF +7/-5%.
Page Name = 67_RF_MMPA
REV: V10
Design Name = S96116-1-13_MB_20190409
DEPARTMENT = WINGTECH
DESIGNER = LIUFENGLEI
Page Modify Date = Monday, April 08, 2019
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D D

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A A

Title			
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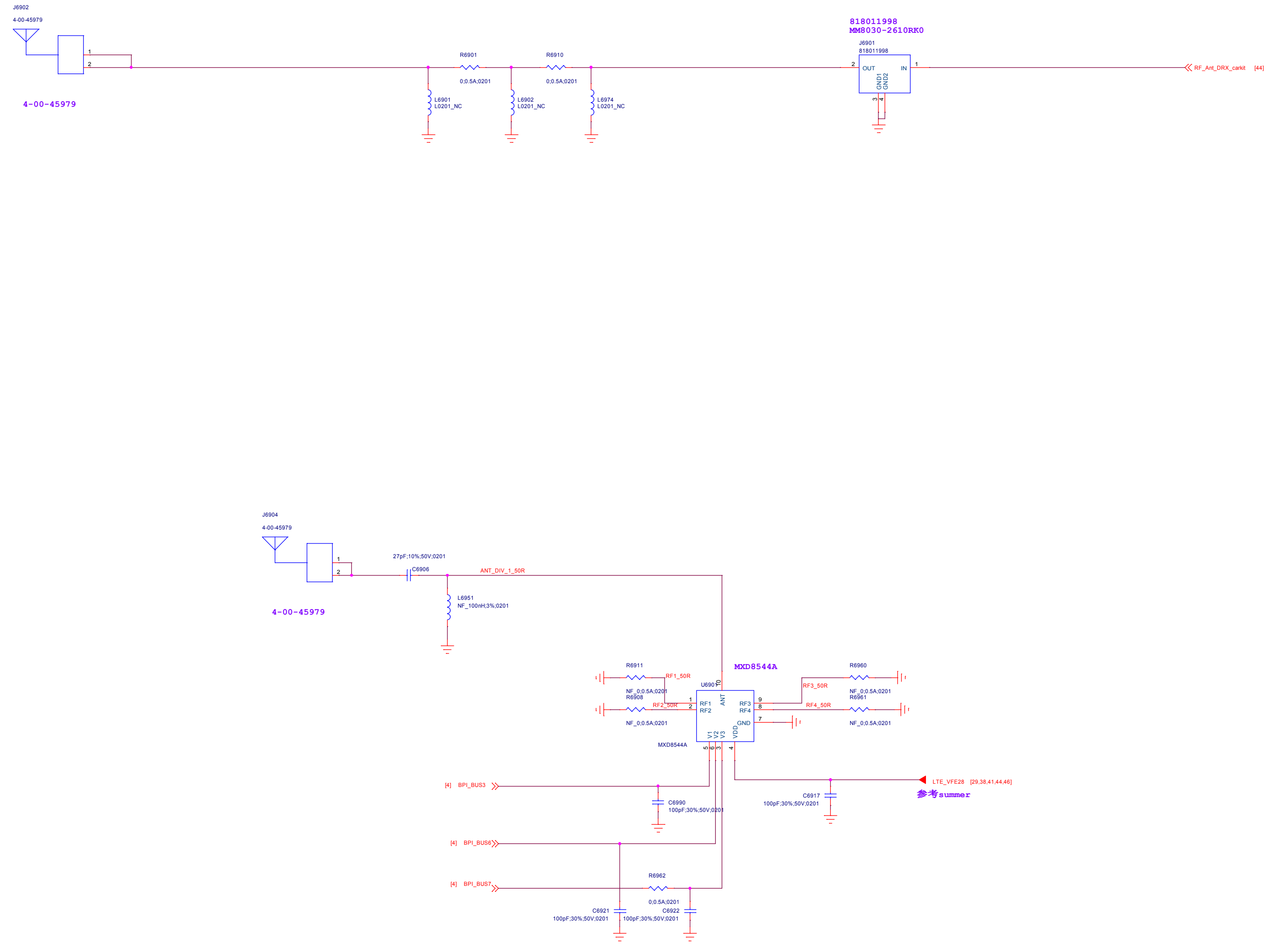
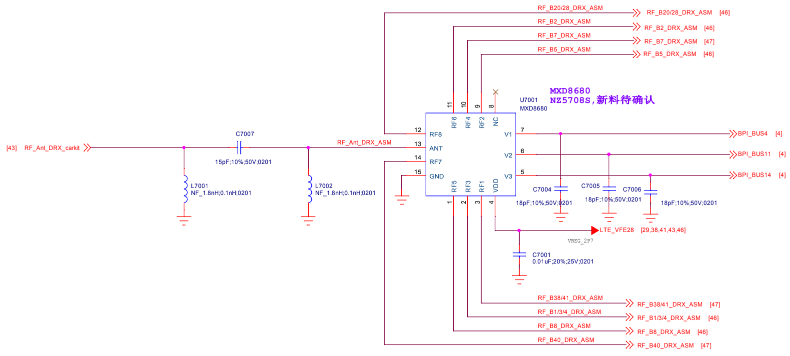
Size A1	Document Number <Doc>		Rev 1.0
Date: Monday, April 08, 2019	Sheet 42 of 53		

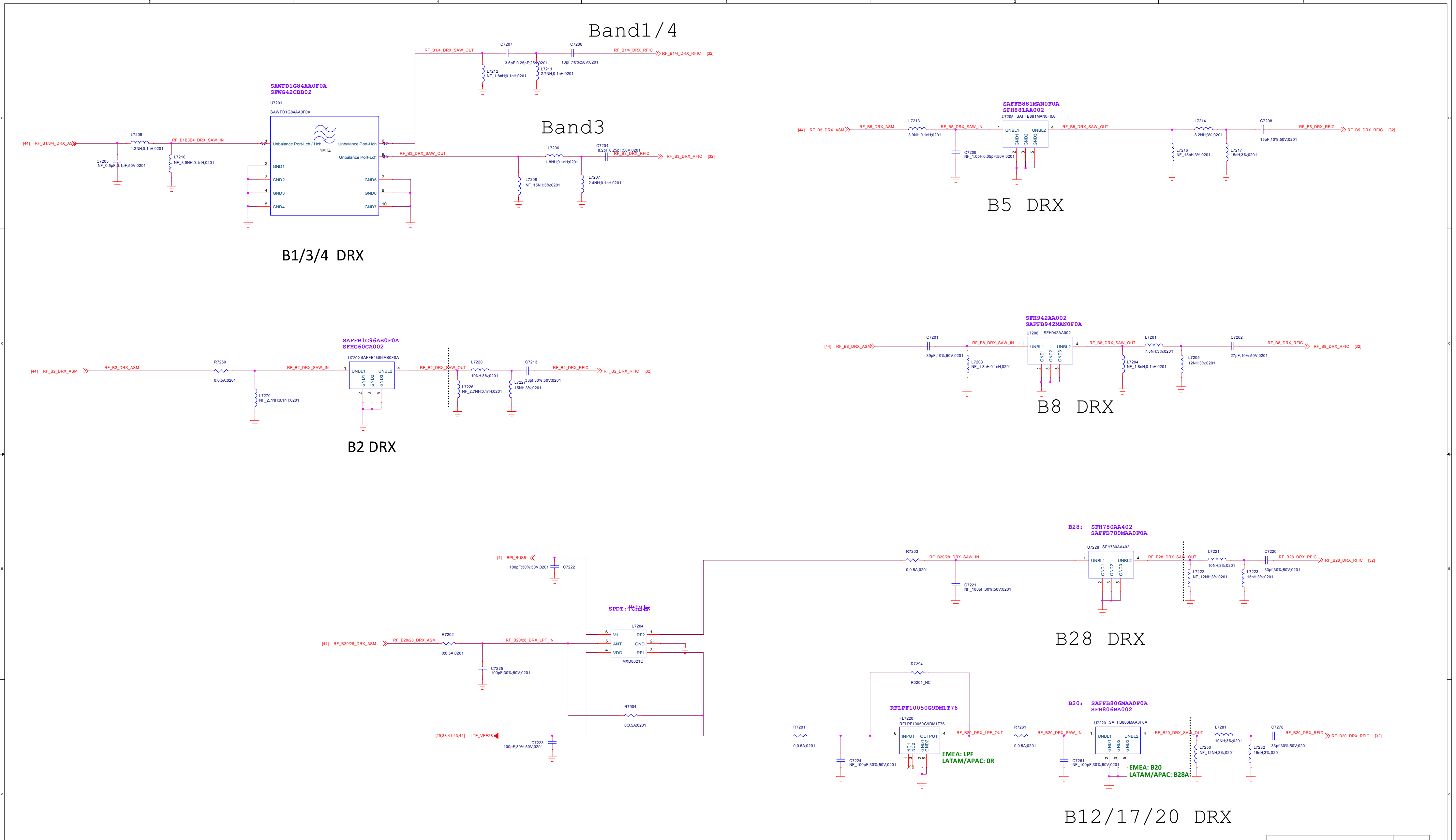


Title Page Name = 69_Diversity_ANT	REV: V10
DOCUMENT NO.:Design Name = S96116-1-13_MB_20190409-1317	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER:DESIGNER = LIUFENGLEI
Date:Page Modify Date = Monday, April 08, 2019	Sheet 43 of 53



Title	Page Name = 70_Diversity ASM — REV V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1317 — Size D
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ — DESIGNER: DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019 — Sheet 44 of 53

Title	Reserved		REV: V10
DOCUMENT NO.:	Design Name		Size A1
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 45 of 53	



Title	Page Name = 72_DRX_LMB
REV:	V10
DOCUMENT NO:	Design Name = S96116-1-13_MB_20190408
DEPARTMENT:	DEPARTMENT = WINGTECH-SZ
DESIGNER:	DESIGNER = LIUFENGLEI
Date:	Page Modify Date = Monday, April 08, 2019
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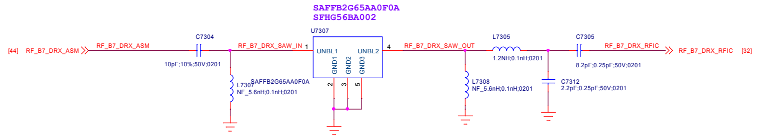
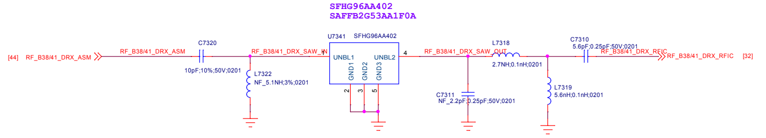
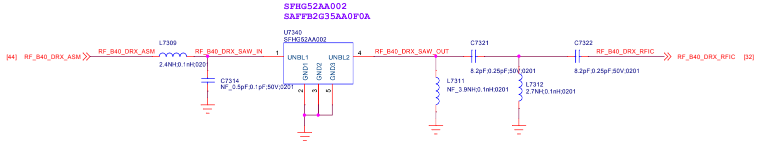


B7 DRX



B38/41 DRX



B40 DRX

Title	Page Name = 73_DRX_HB	REV: V10
DOCUMENT NO.:	Design Name = S96116-1-13_MB_20190409-1317	Size D
DEPARTMENT: DEPARTMENT = WINGTECH-SZ	DESIGNER: DESIGNER = LIUFENGLEI	
Date: Page Modify Date = Monday, April 08, 2019	Sheet 47 of 53	

5	4	3	2	1

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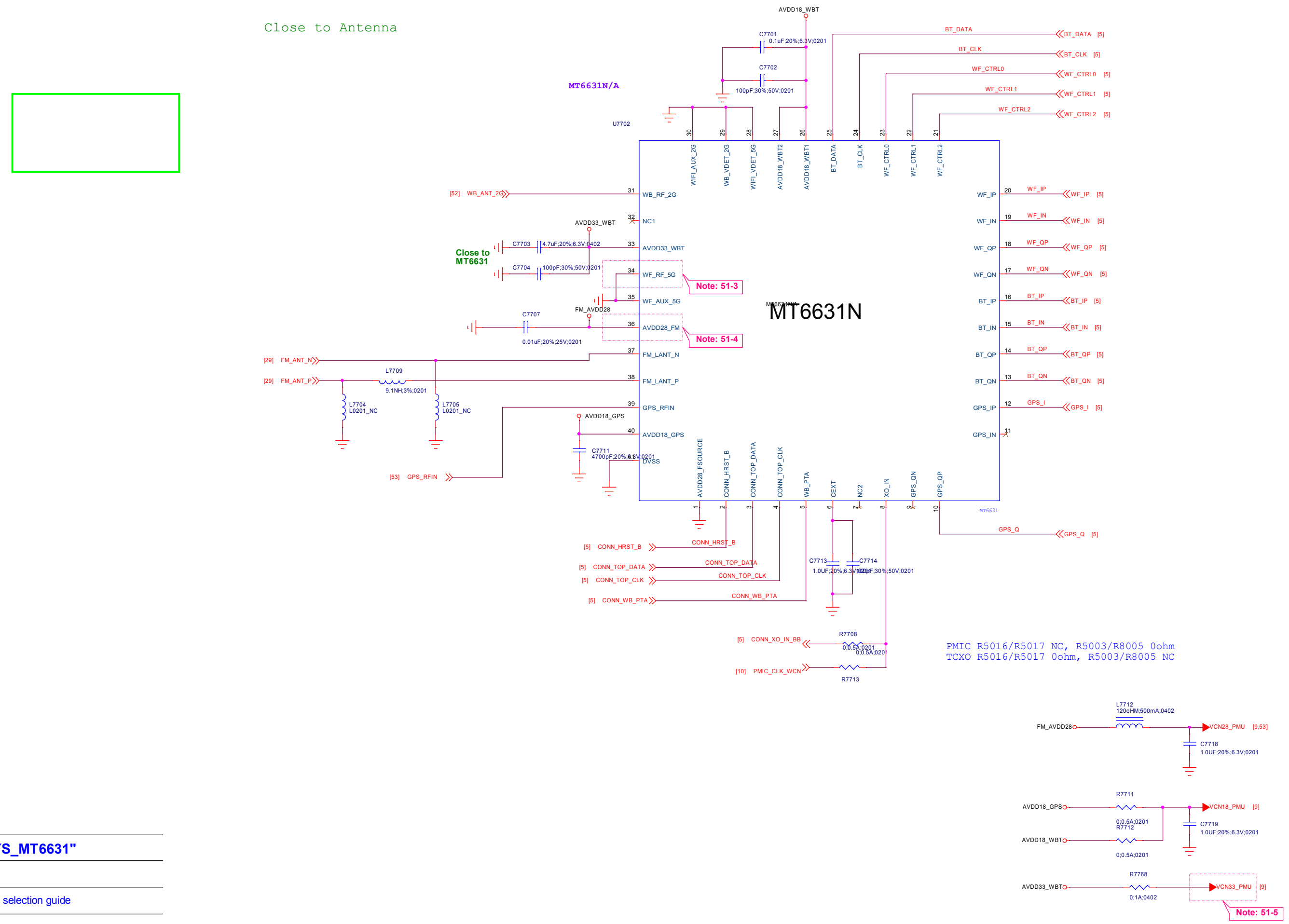
Title	<Title>					
Size A1	Document Number <Doc>					Rev 1.0
Date:	Monday, April 08, 2019	Sheet	48	of	53	

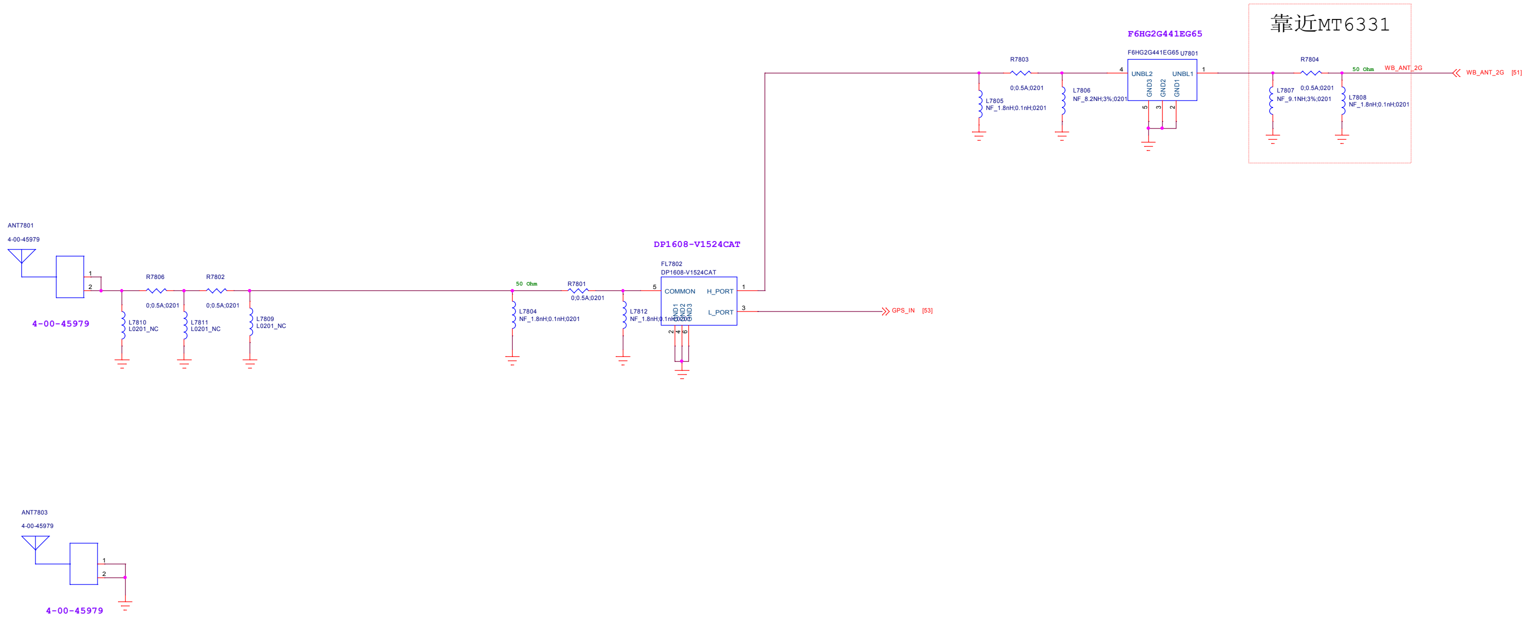
5 4 3 2 1

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 49 of 53	

Title	Reserved	REV: V10	
DOCUMENT NO.:	Design Name	Size A1	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
Date:	Monday, April 08, 2019	Sheet 50 of 53	



Schematic design notice of "51_CONNECTIVITY_CONSYS_MT6631"	
Note 51-1:	For R5015 size, please select 0402 size or larger one
Note 51-2:	Please refer to MT6762 Baseband design notice for VCN33 LDO selection guide
Note 51-3:	If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND
Note 51-4:	Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support
Note 51-5:	If WiFi 5G were no need, VCN33 could be chosen from PMIC output (VCN33_PMU)



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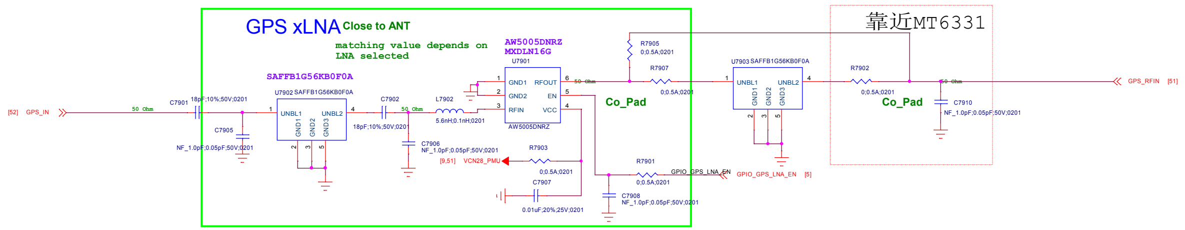
DOCUMENT NO.:	Design Name	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Monday, April 08, 2019	Sheet 52 of 53	

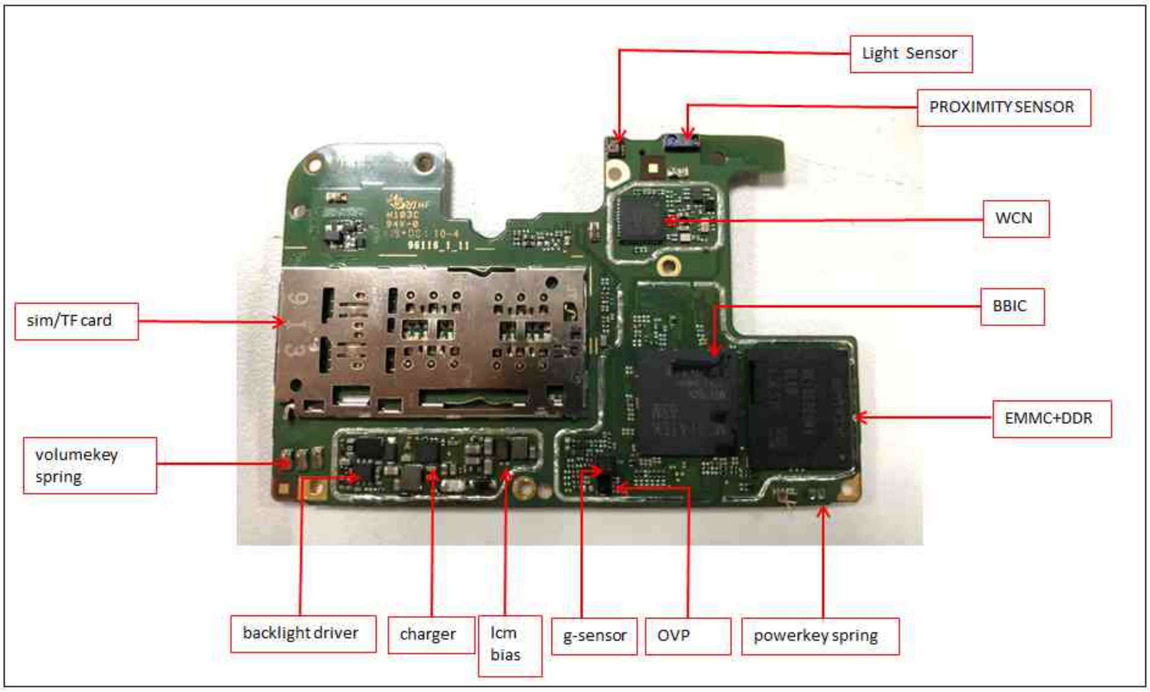


Title	**Reserved**	REV: V10	
DOCUMENT NO.:	**Design Name**	Size D	
DEPARTMENT:	WINGTECH-SH	DESIGNER:	LIUFENGLEI
WINGTECH			
Date:	Monday, April 08, 2019	Sheet 53 of 53	

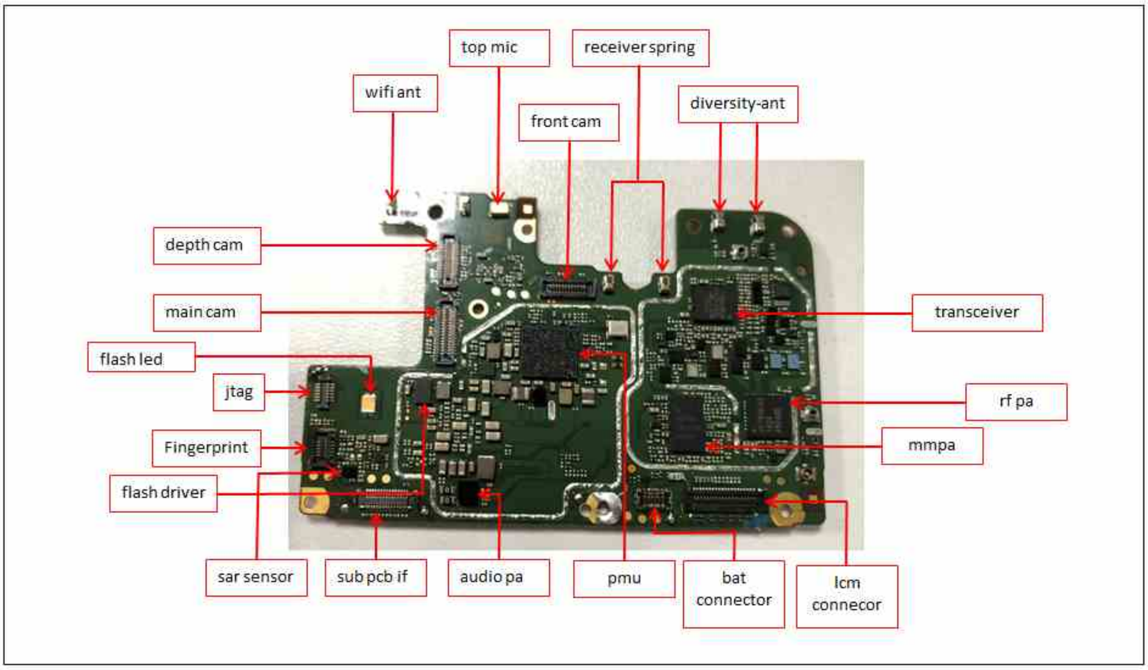
8. Level 3 Repair

8-1. Components Layout

Main board top view

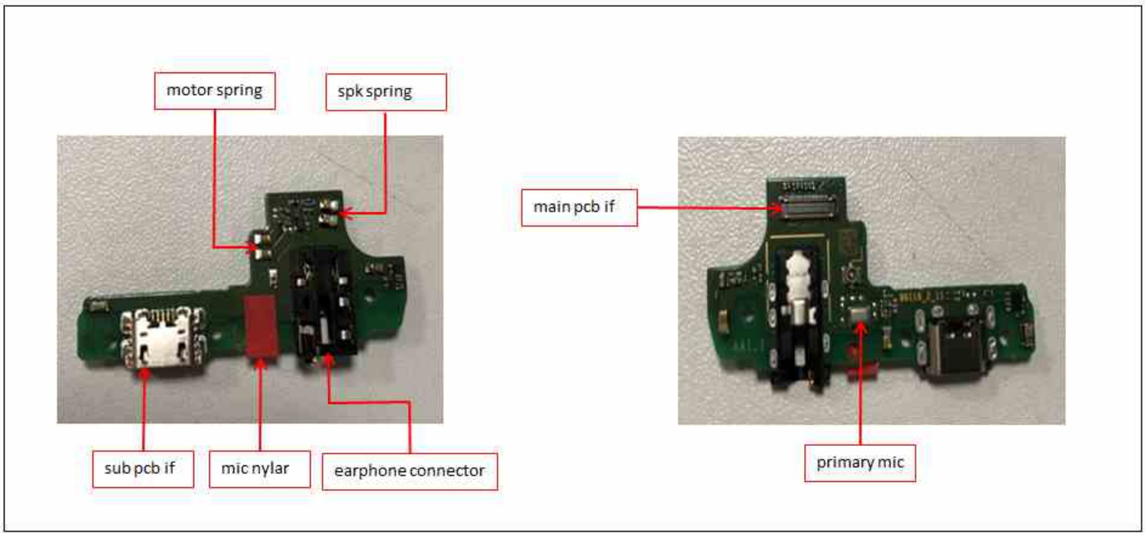


Main board bot view

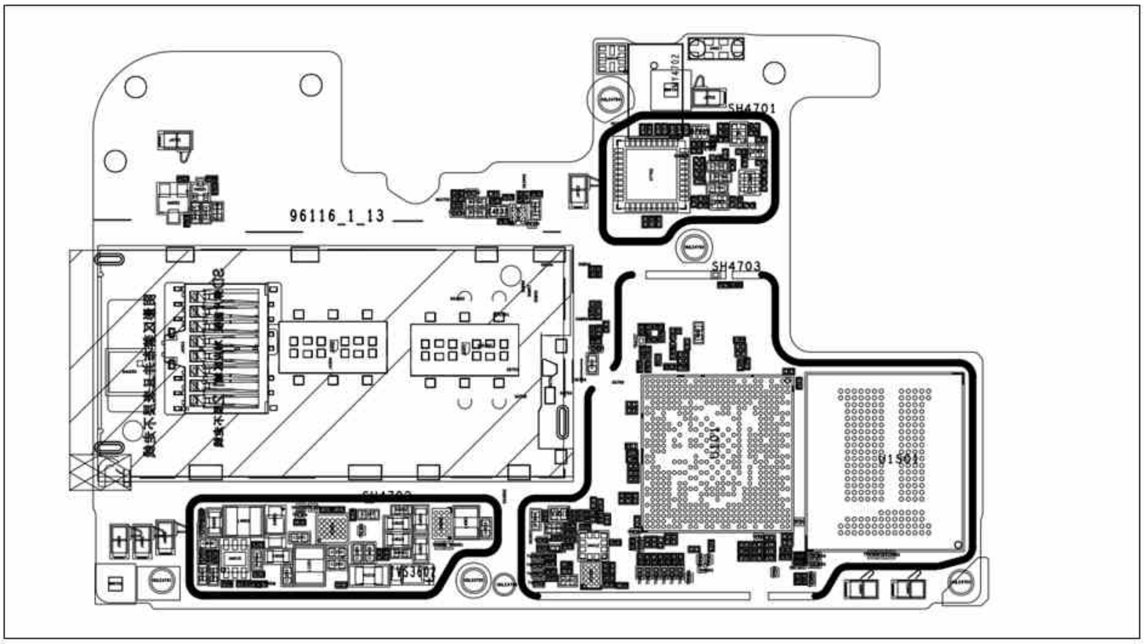


8. Level 3 Repair

Sub board

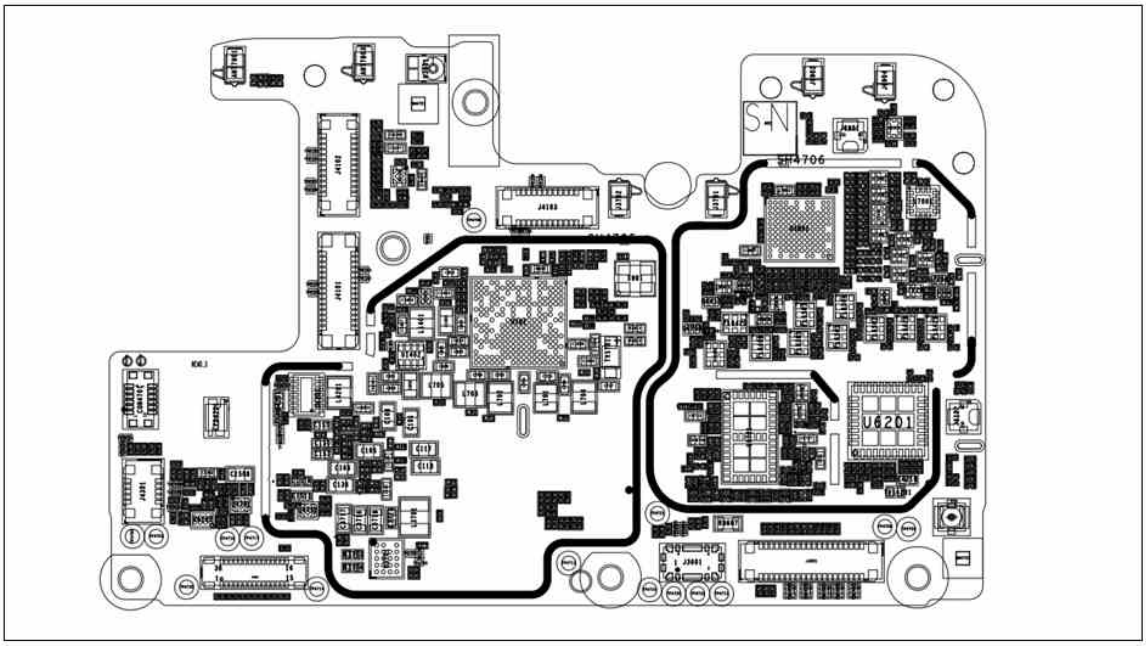


Reference map
Mb top view

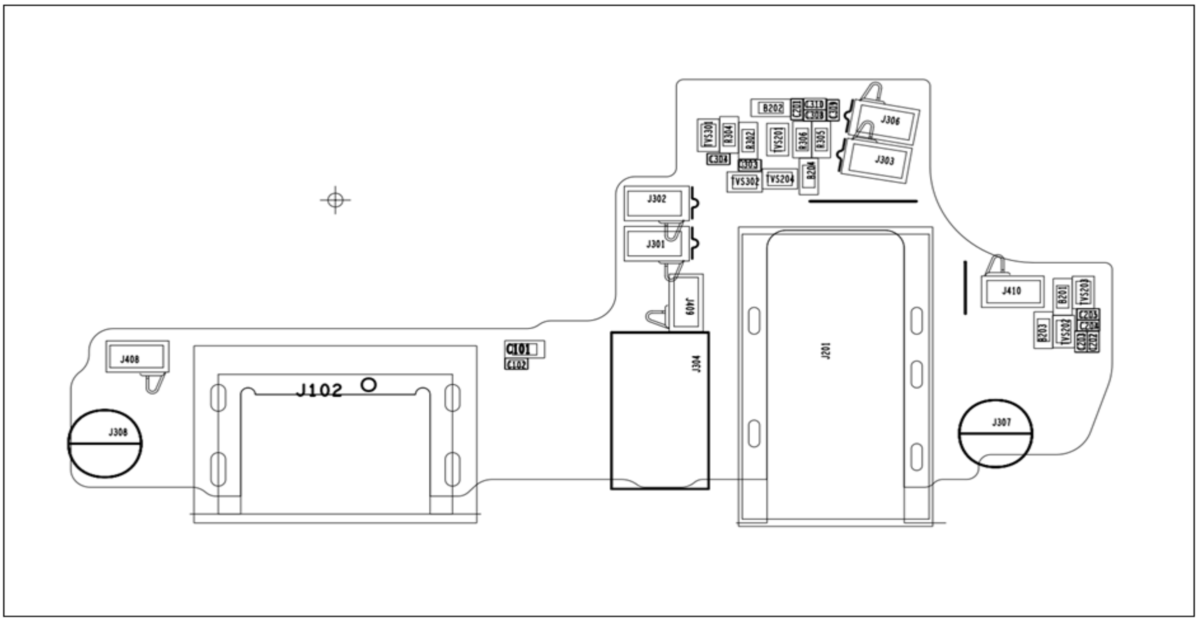


8. Level 3 Repair

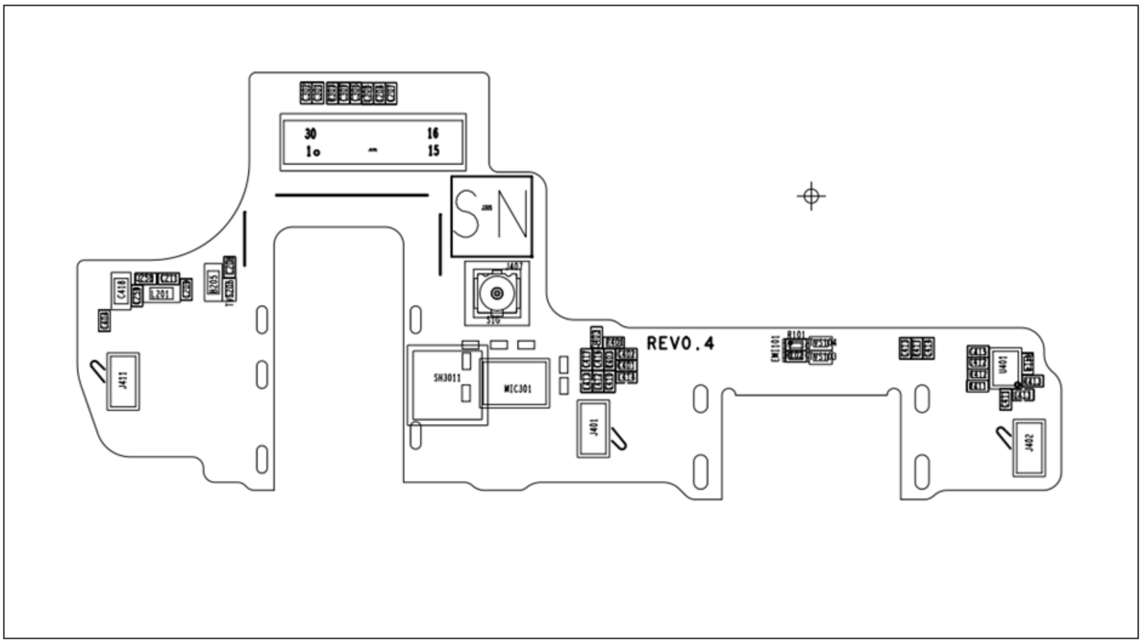
Mb bot view

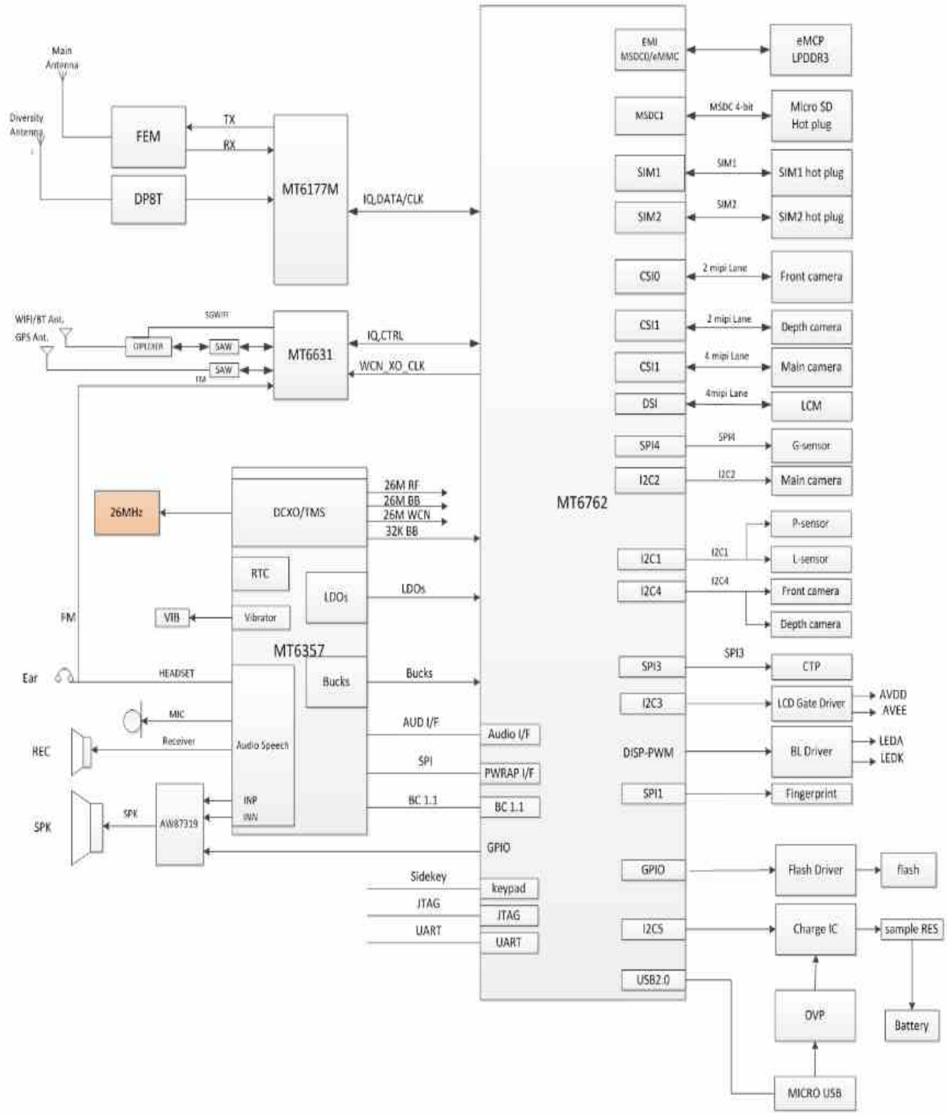


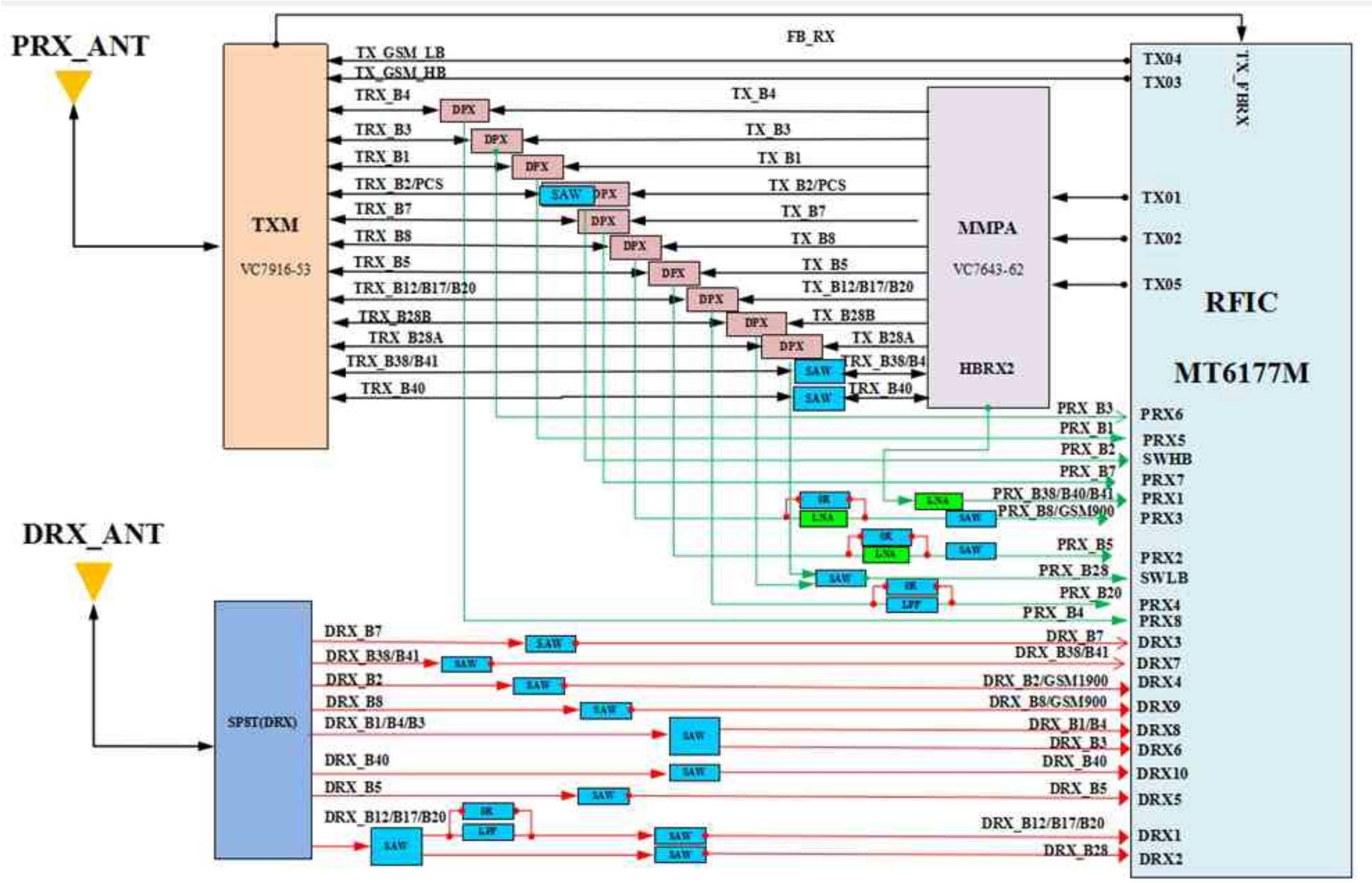
Sub top view



Sub bot view

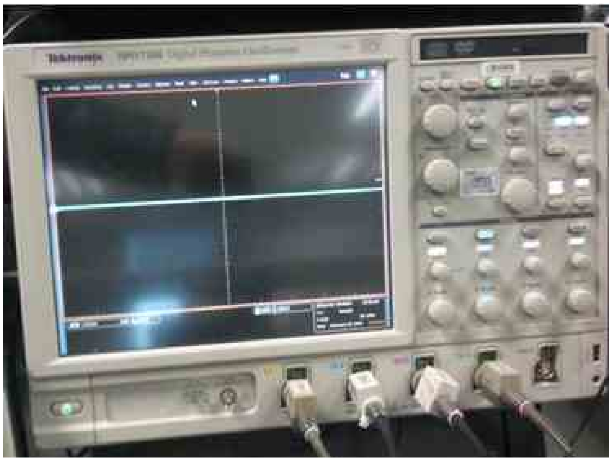
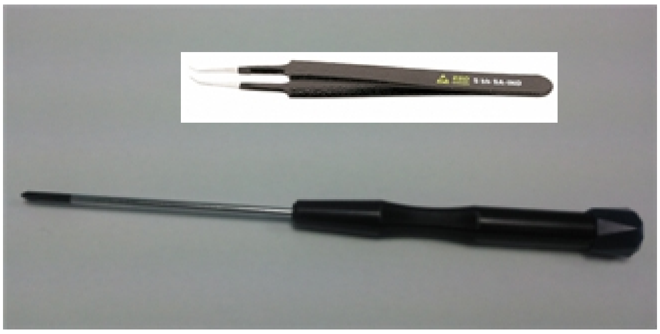
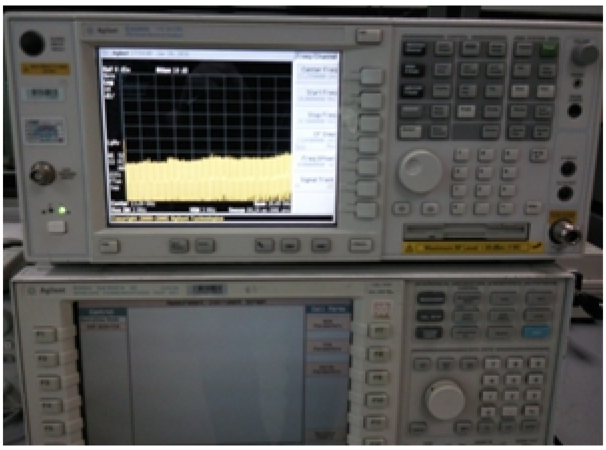



Main Antenna
Diversity Antenna
FEM
DPBT
TX
RX
MT6177M
IQ_DATA/CLK
WIFI/BT Ant.
GPS Ant.
SAW
SAW
MT6631
IQ_CTRL
WCN_XO_CLK
26MHz
DCXO/TMS
26M RF
26M BB
26M WCN
32K BB
RTC
LDOs
LDOs
VIB
Vibrator
MT6357
FM
Ear
HEADSET
Bucks
Bucks
MIC
AUD I/F
Receiver
Audio Speech
REC
SPI
SPK
SPK
AW87319
INP
INN
BC 1.1
GPIO
Sidekey
JTAG
UART
Audio I/F
PWRAP I/F
BC 1.1
keypad
JTAG
UART
MT6762
EMI MSDC0/eMMC
eMCP LPDDR3
MSDC1
MSDC 4-bit
Micro SD Hot plug
SIM1
SIM1
SIM1 hot plug
SIM2
SIM2
SIM2 hot plug
CSI0
2 mipi Lane
Front camera
CSI1
2 mipi Lane
Depth camera
CSI1
4 mipi Lane
Main camera
DSI
4mipi Lane
LCM
SPI4
SPI4
G-sensor
I2C2
I2C2
Main camera
P-sensor
I2C1
I2C1
L-sensor
I2C4
I2C4
Front camera
Depth camera
SPI3
SPI3
CTP
I2C3
LCD Gate Driver
AVDD
AVEE
DISP-PWM
BL Driver
LEDA
LEDK
SPI1
Fingerprint
GPIO
Flash Driver
flash
I2C5
Charge IC
sample RES
Battery
USB2.0
OVP
MICRO USB



PRX_ANT
DRX_ANT
FB_RX
TXM
VC7916-53
MMPA
VC7643-62
HBRX2
RFIC
MT6177M
SP8T(DRX)
TX_GSM_LB
TX_GSM_HB
TRX_B4
TRX_B3
TRX_B1
TRX_B2/PCS
TRX_B7
TRX_B8
TRX_B5
TRX_B12/B17/B20
TRX_B28B
TRX_B28A
TRX_B38/B41
TRX_B40
TX_B4
TX_B3
TX_B1
TX_B2/PCS
TX_B7
TX_B8
TX_B5
TX_B12/B17/B20
TX_B28B
TX_B28A
TX04
TX03
TX01
TX02
TX05
TX_FBRX
PRX_B3
PRX_B1
PRX_B2
PRX_B7
PRX_B38/B40/B41
PRX_B8/GSM900
PRX_B5
PRX_B28
PRX_B20
PRX_B4
PRX6
PRX5
SWHB
PRX7
PRX1
PRX3
PRX2
SWLB
PRX4
PRX8
DRX_B7
DRX_B38/B41
DRX_B2
DRX_B8
DRX_B1/B4/B3
DRX_B40
DRX_B5
DRX_B12/B17/B20
DRX_B2/GSM1900
DRX_B8/GSM900
DRX_B1/B4
DRX_B3
DRX_B28
DRX3
DRX7
DRX4
DRX9
DRX8
DRX6
DRX10
DRX5
DRX1
DRX2
DPX
SAW
LNA
LPF

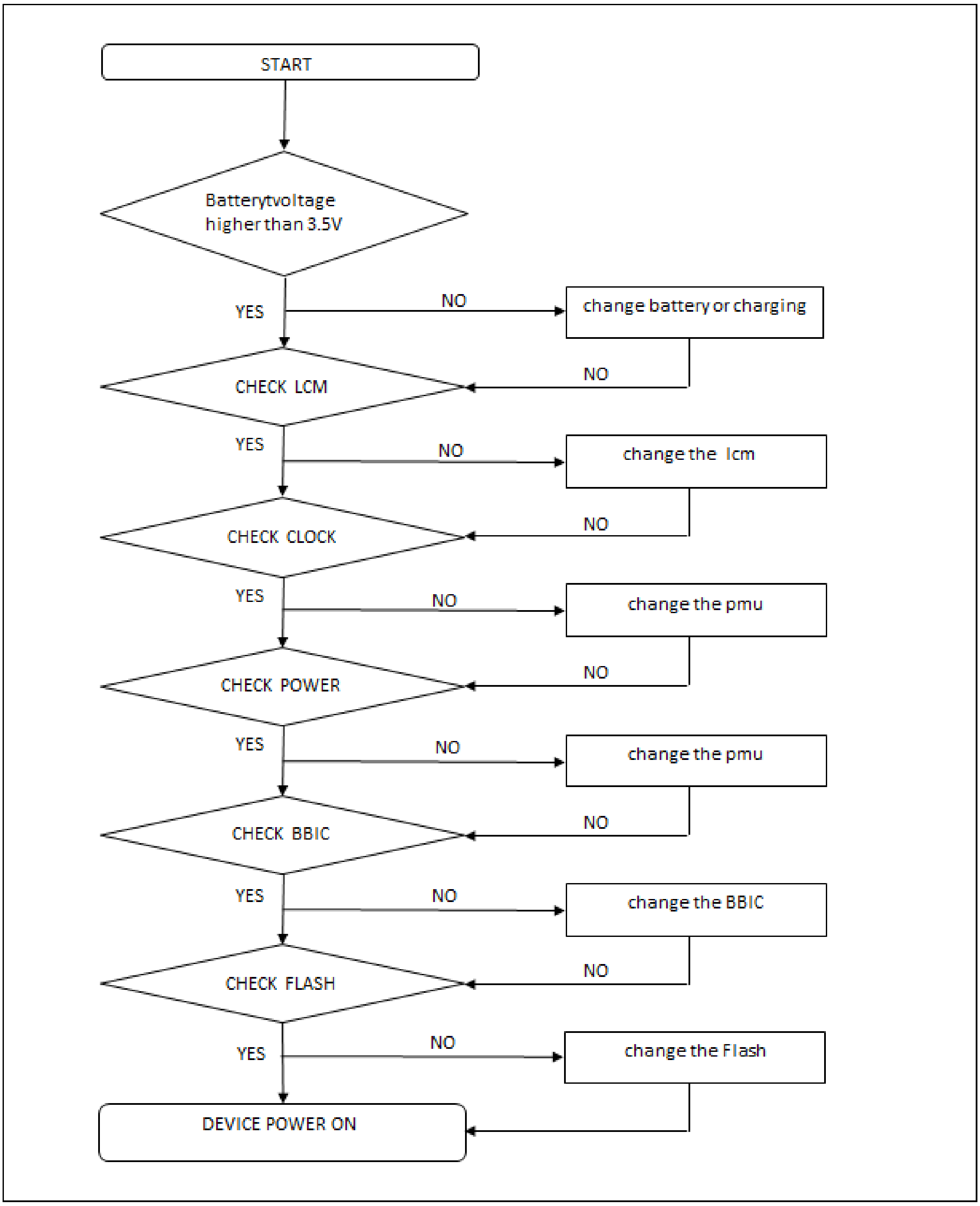
8-3. Flow chart of Troubleshooting.

	
Oscilloscope	**Digital multimeter**
	
Power Supply	**+ driver, ESD Safe Tweezer**
	
8960 & Spectrum Analyzer	**Soldering iron**

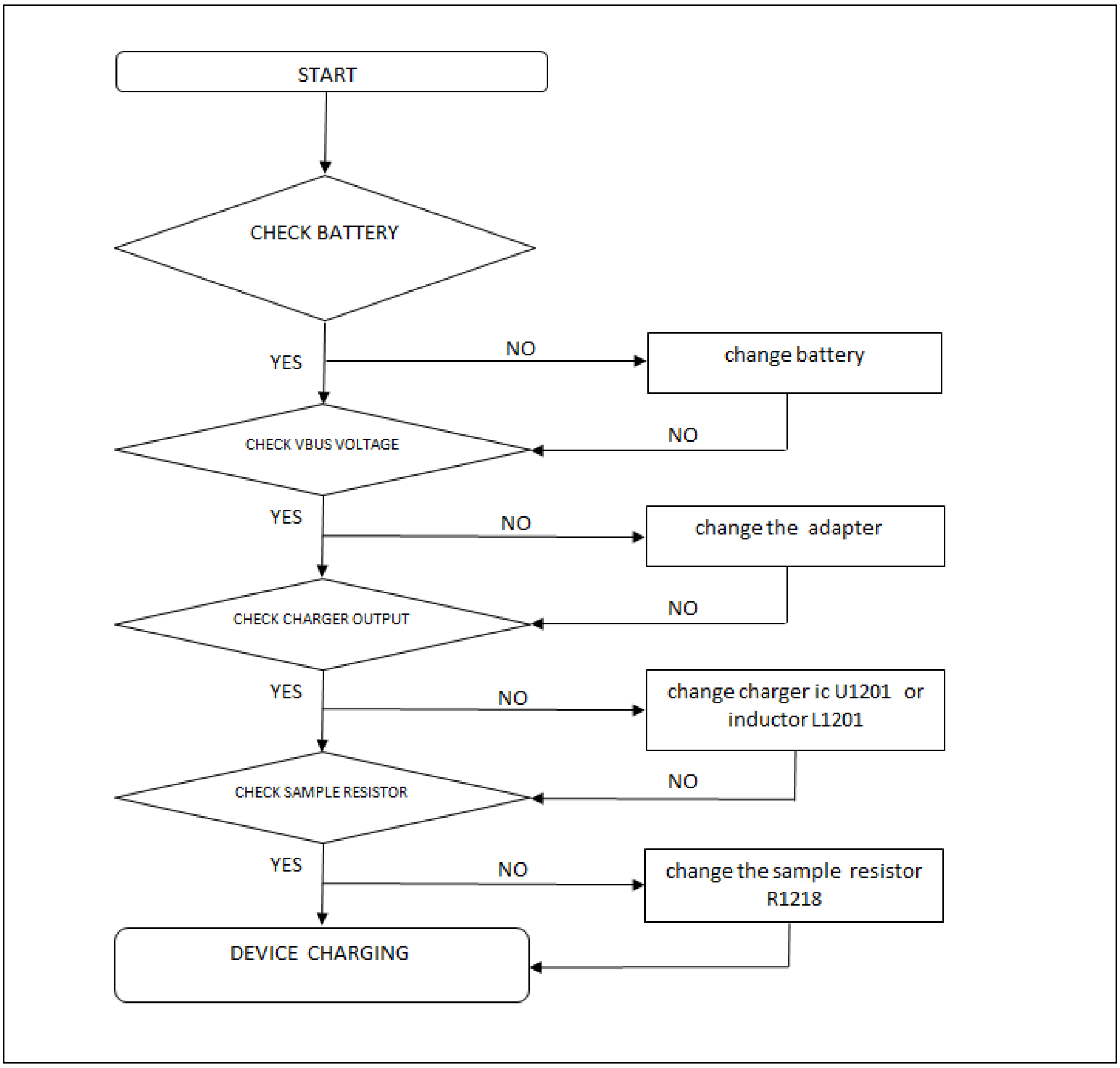
8. Level 3 Repair

8-4-1. Power On

: Checking Power signal (LCM, PMU, Clock)

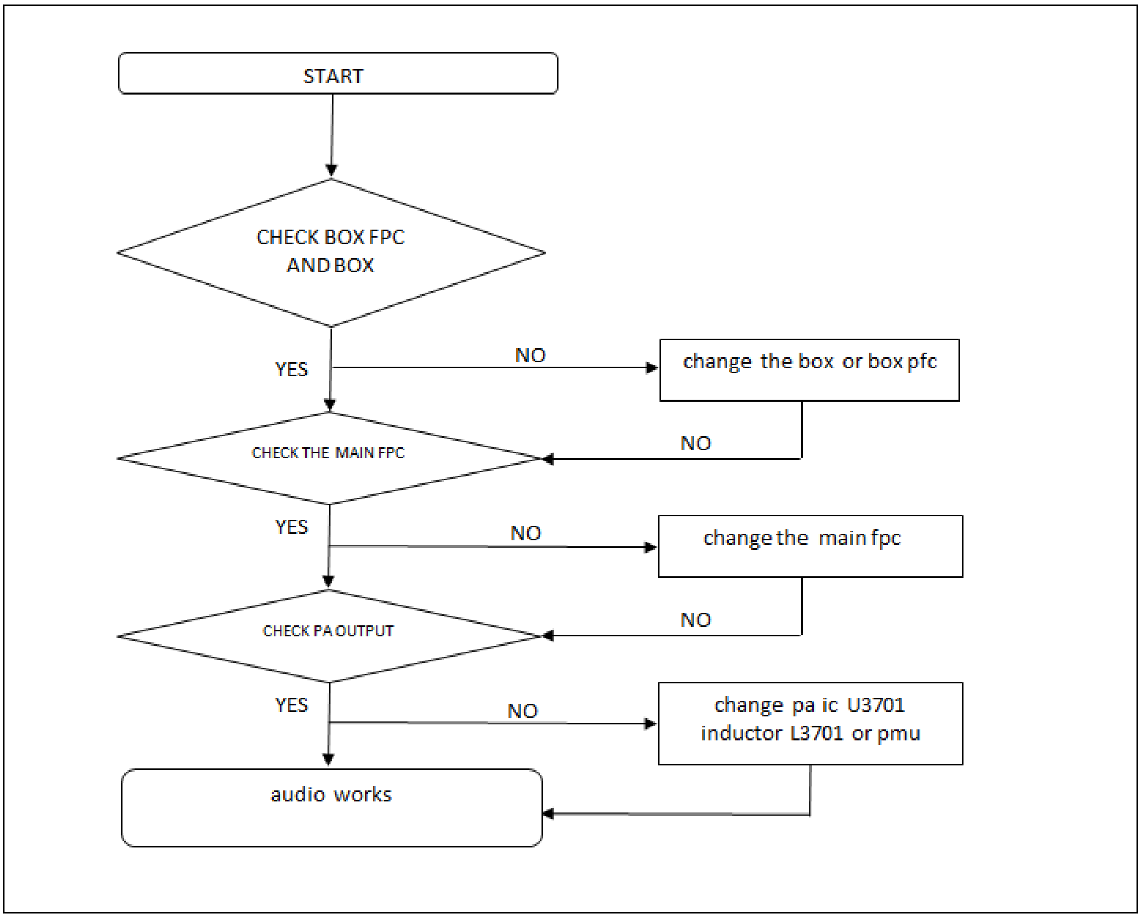


8-4-2. Charging
: The charging controlled by third charger chip



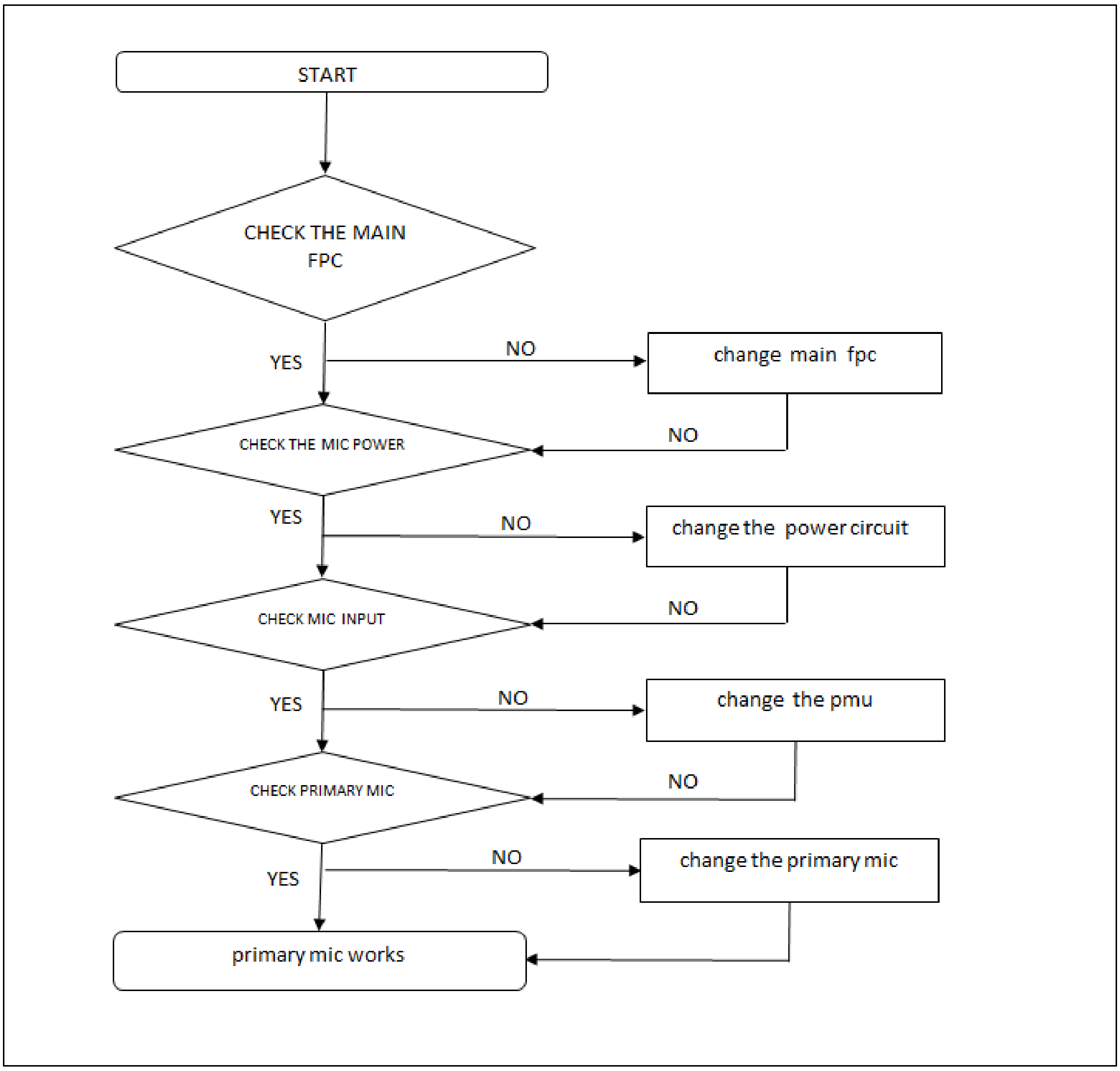
8-4-3. Audio speaker

: The Speaker control signals are generated by MT6357 (U701) and PA AW87319 (U3701), the chip, PA and the speaker are to be checked out.



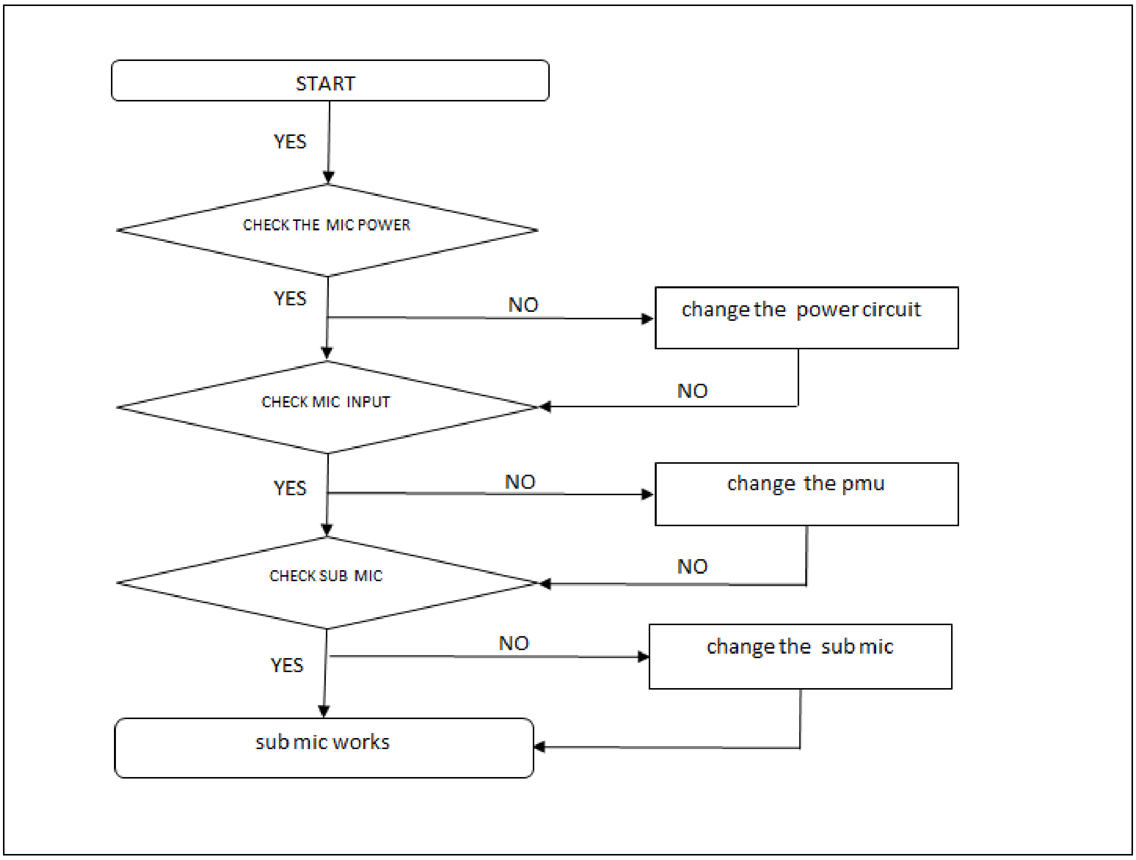
8-4-4. Audio Main MIC

: The MIC control signals are generated by PMU chip MT6357 (U701), the PMU chip and the MIC are to be checked out.



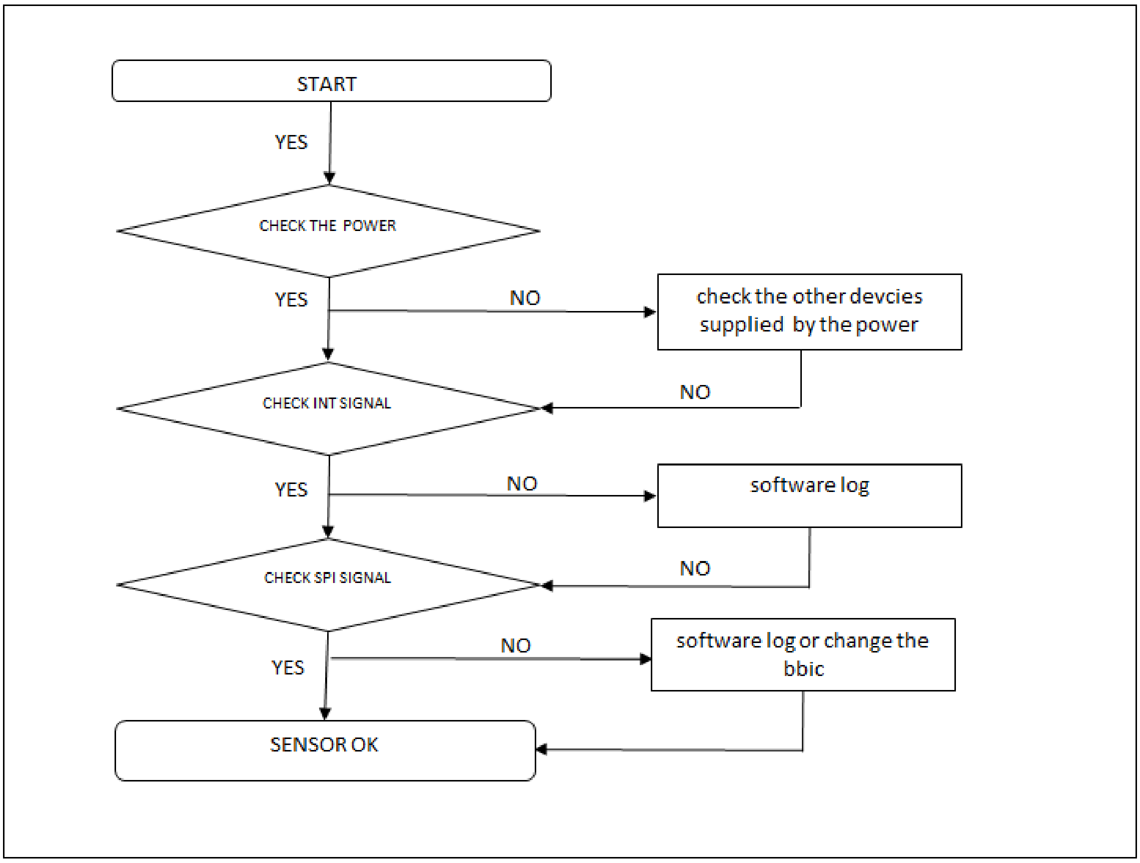
8-4-5. Sub MIC

: The MIC control signals are generated by PMU chip MT6357 (U701), the PMU chip and the MIC are to be checked out.



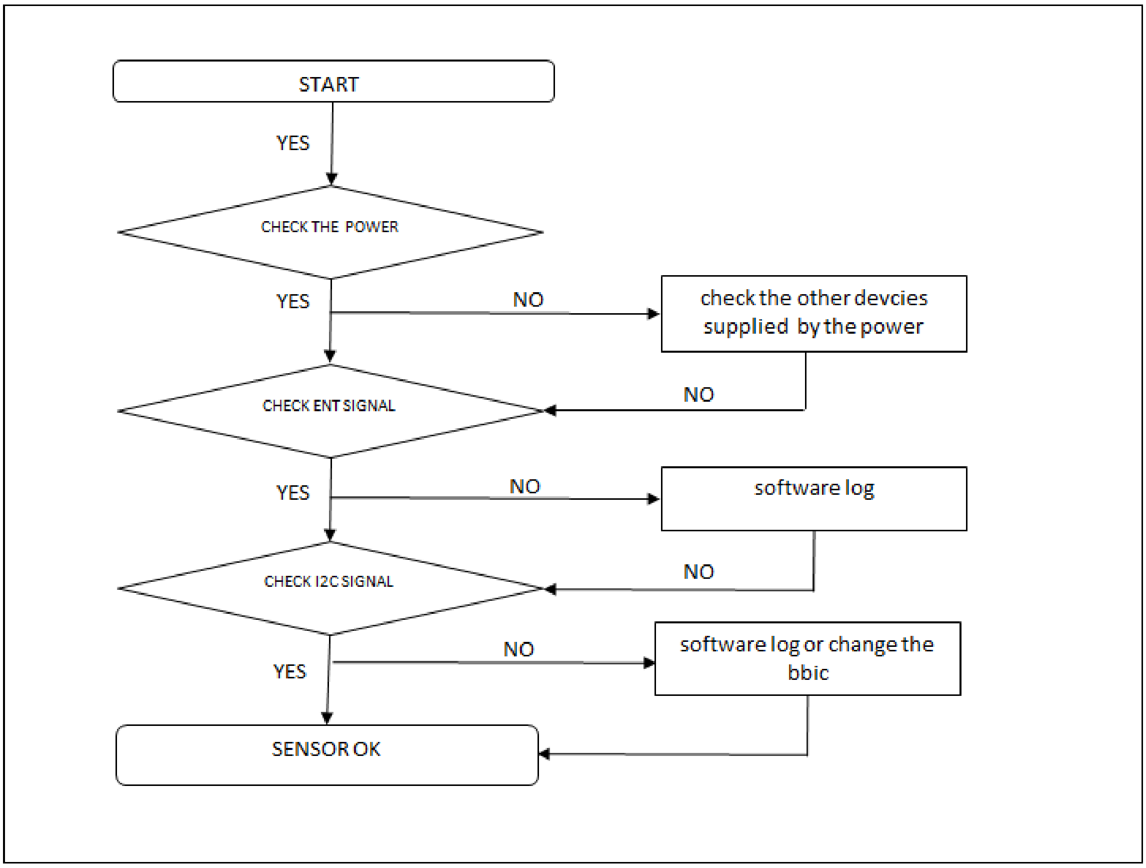
8-4-6. Accelerometer sensor

: The Accelerometer sensor is calibrated by using SW algorithm.



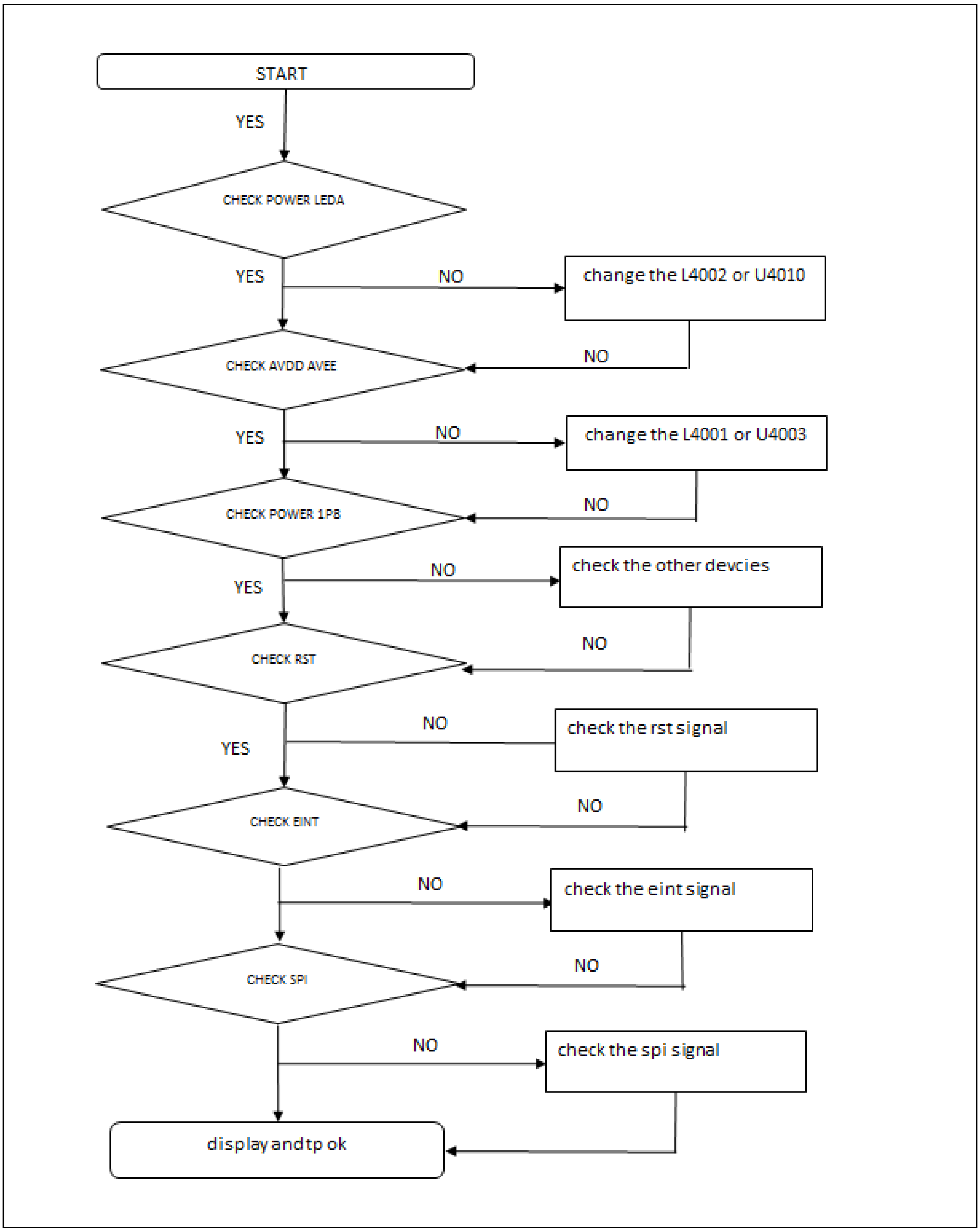
8-4-7. Proximity and light sensor

: Proximity (U4417) and Light Sensor (U4416) is worked as below: Control the screen's on/off operation automatically while making phone calls, and adjust the screen brightness according to ambient light.



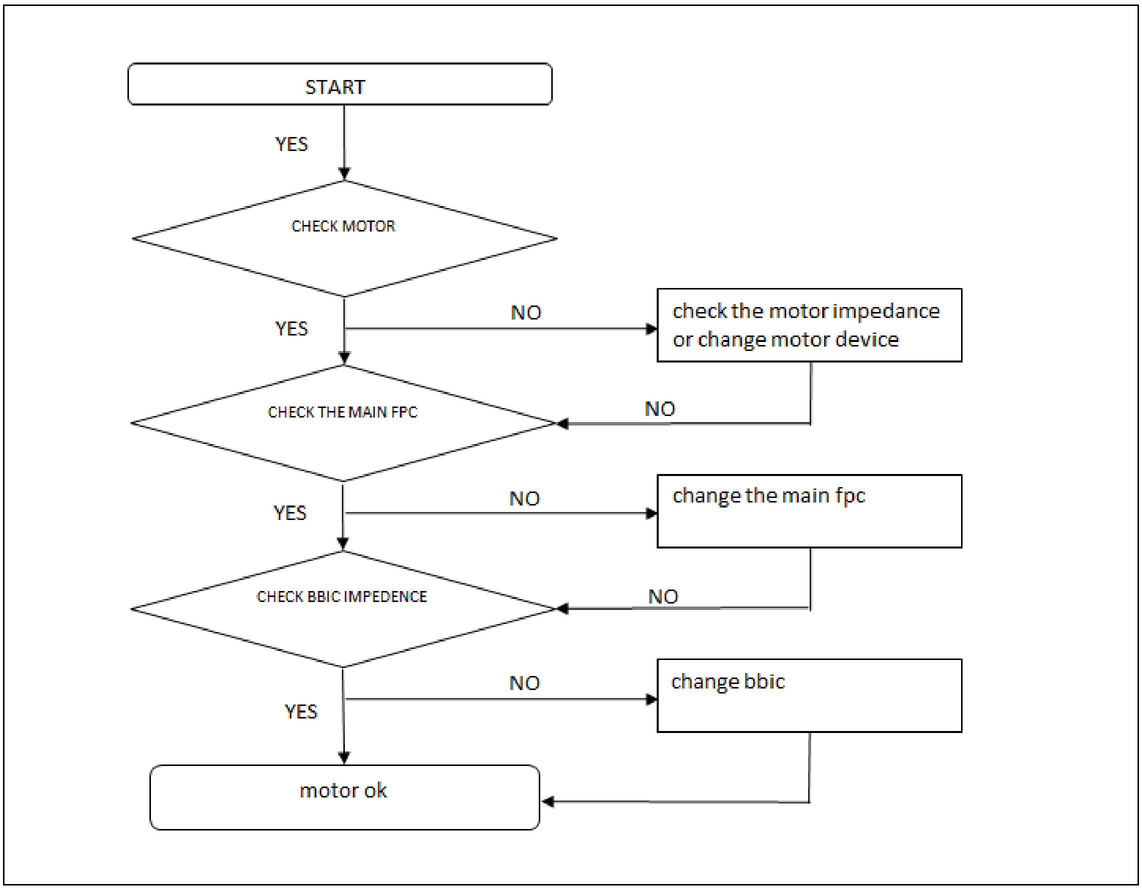
8-4-8. TOUCH SCREEN AND DISPLAY

: The Touch control signals are generated by MT6762. It is assembled with LCD.



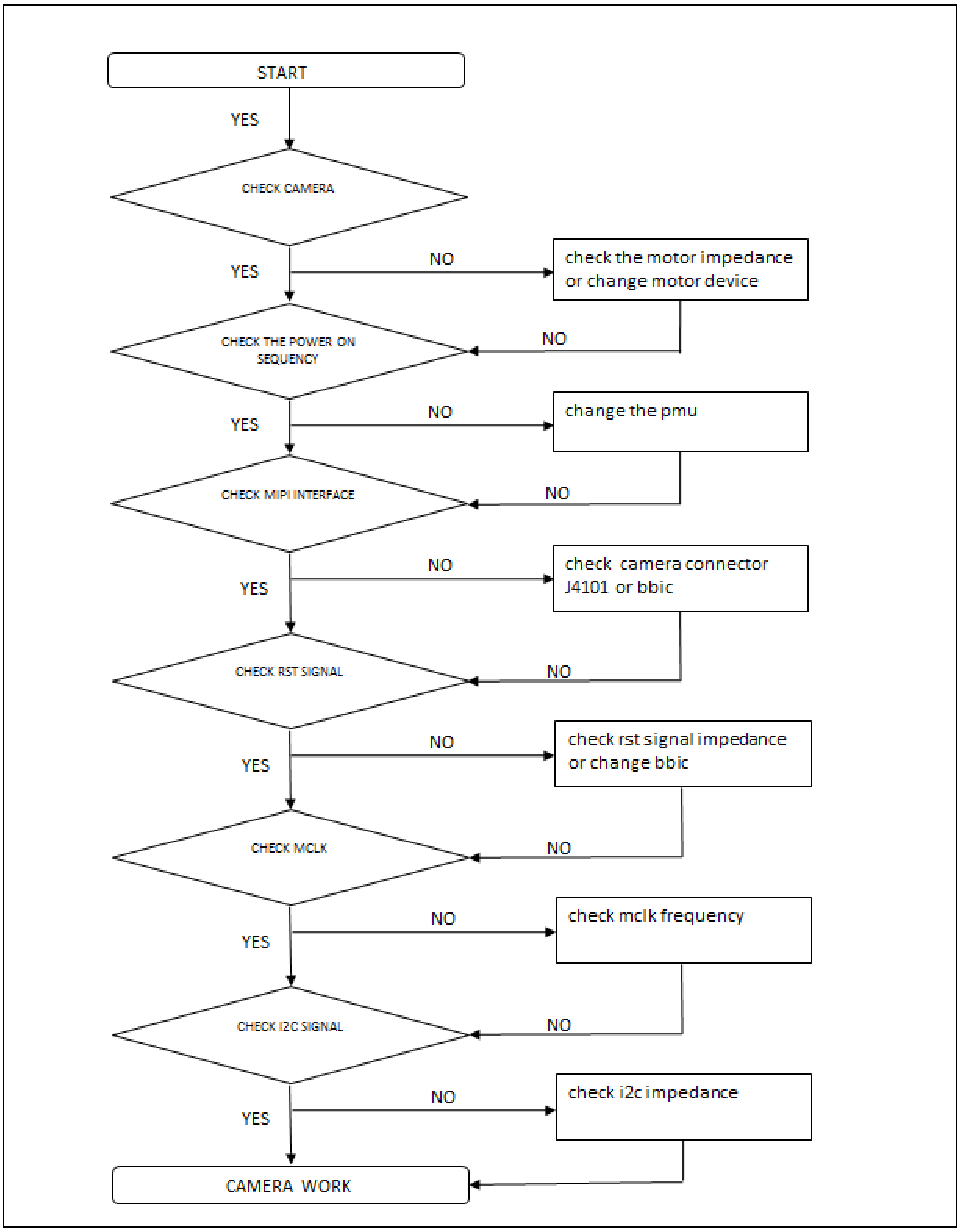
8-4-9. Vibrator

: The Vibrator control signals are generated by MT6357.



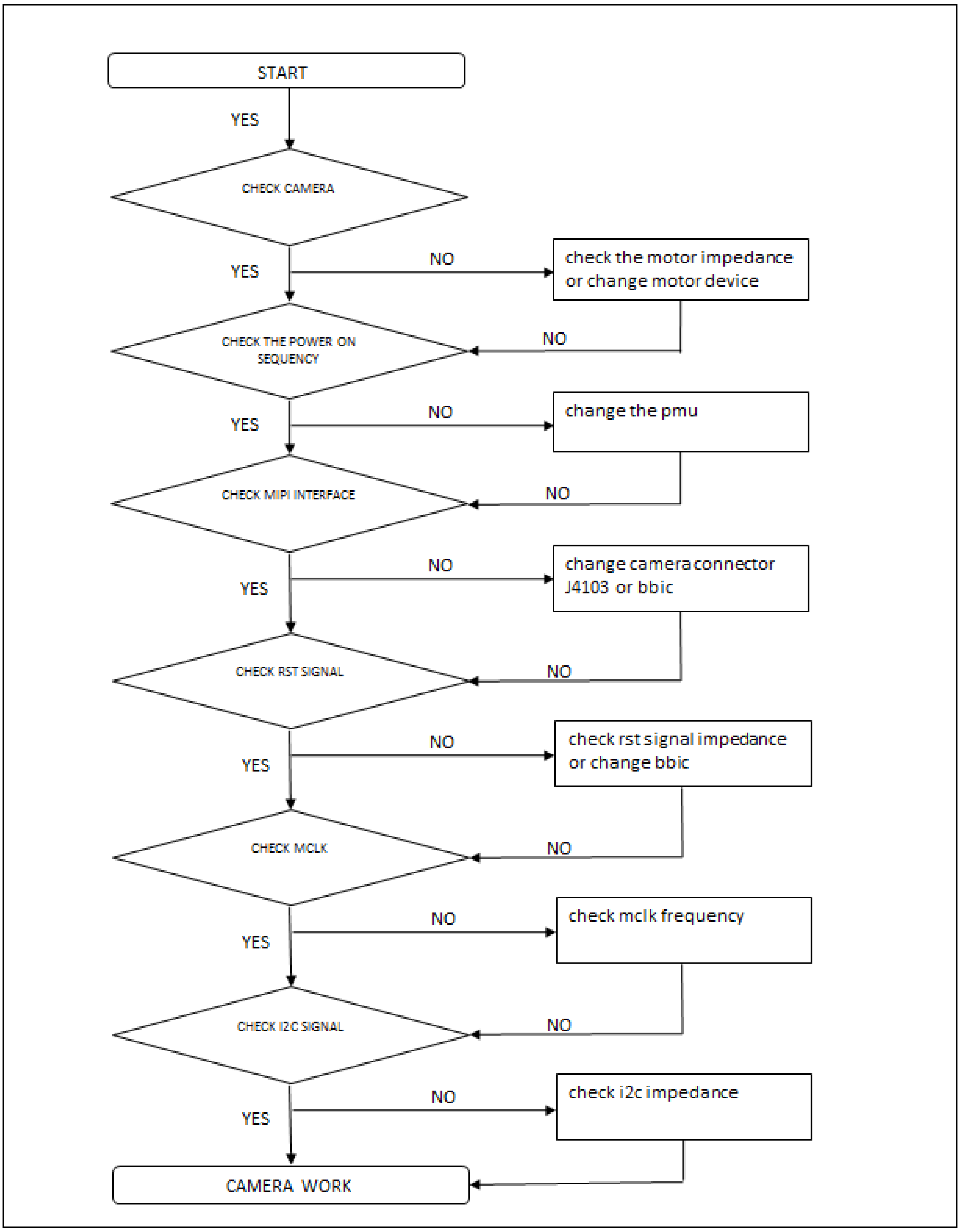
8-4-10. Rear Camera

: The Camera control signals are generated by MT6762.



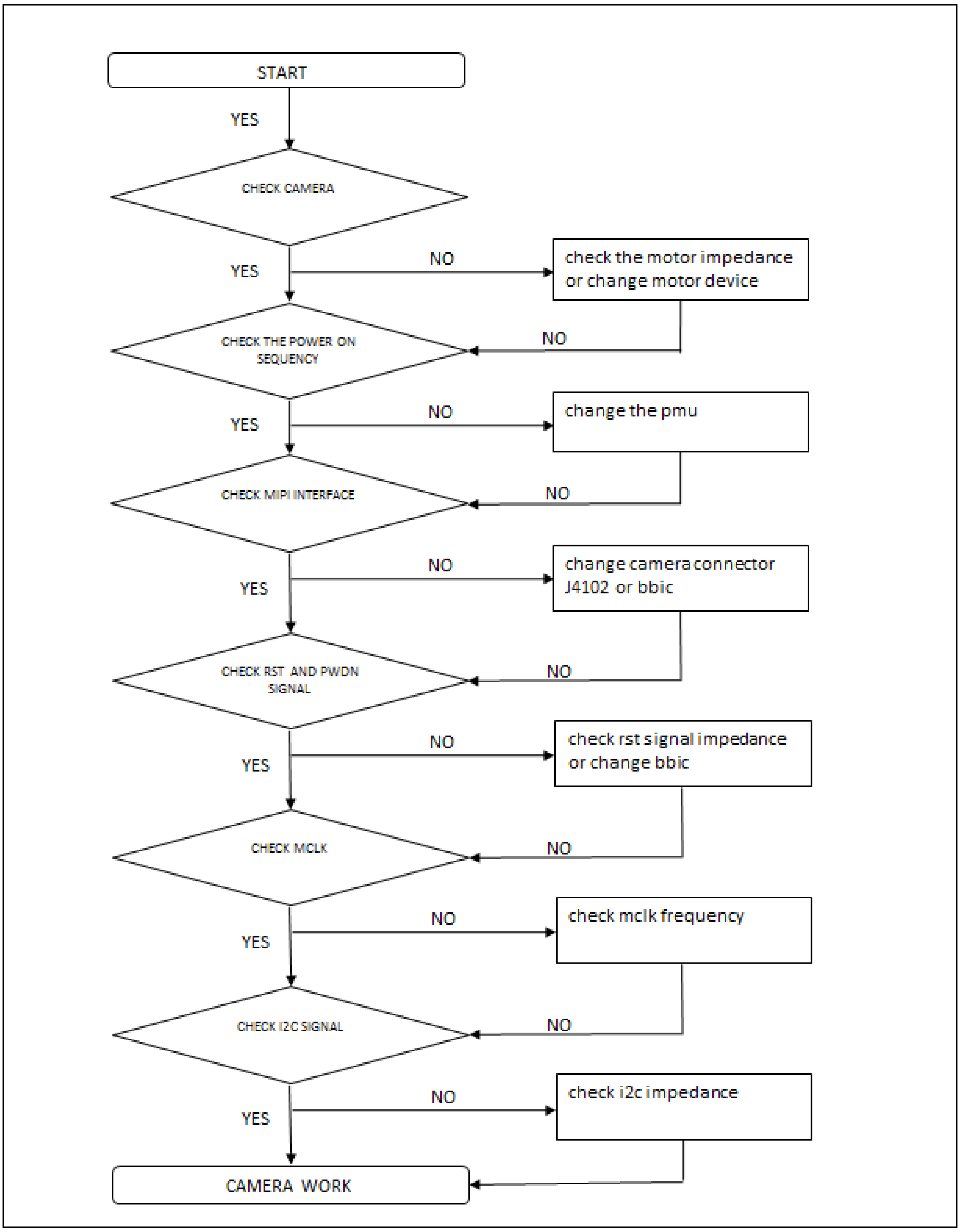
8-4-11. Front Camera

The Camera control signals are generated by MT6762.



8-4-12. Depth Camera

: The depth Camera control signals are generated by MT6762.



8-5. Service Schematics

- U101_MT6762_BB chip IC , Digital *Baseband Processor*(Top)

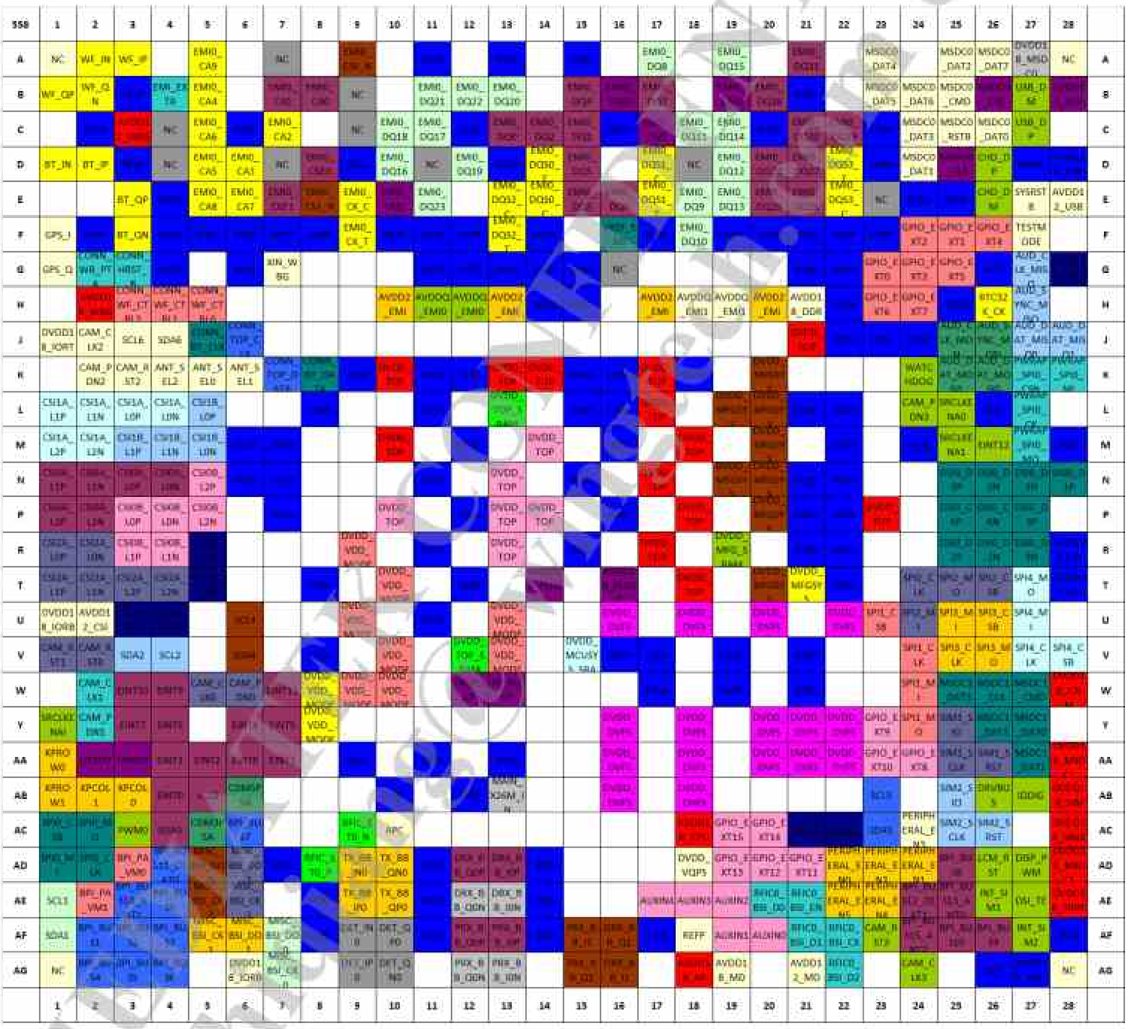


Figure 2-1. Ball map view